Marathwada Mitra Mandal's

COLLEGE OF ENGINEERING, PUNE

An autonomous Institute Affiliated to SPPU

'येथे बहुतांचे हित ।'

Department of Computer Engineering

Lab Manual

310256 : Data Science And Big Data Analytics Laboratory

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TE COMP (2019 Pattern)

Academic Year 2025-26 Sem II

Preface

Data science and Big data analytics use various tools, techniques, and methodologies to extract insights from data.Data science is a multidisciplinary field that uses scientific methods, processes, algorithms, and systems to extract insights and knowledge from structured and unstructured data.Data science encompasses a wide range of activities, including data exploration, statistical analysis, machine learning, predictive modeling, and more.Big data analytics involves the analysis of large and complex datasets, often beyond the capabilities of traditional data processing applications.

Here are the characteristics and properties of a Data science and Big data analytics:

- Integrates principles from statistics, computer science, mathematics, and domain-specific expertise.
- Encompasses data collection, cleaning, exploration, modeling, evaluation, and interpretation.
- Utilizes statistical and machine learning techniques to predict future trends or outcomes.
- Aims to derive actionable insights to support informed decision-making.
- Focuses on handling large Volume, diverse Variety, and high Velocity of data.
- Utilizes distributed computing frameworks like Apache Hadoop and Apache Spark.

MARATHWADA MITRA MANDAL'S
COLLEGE OF ENGINEERING
An Autonomous Institute | Approved by AICTE New Delhi
Recognised by Directorate of Technical Education Mumbai
Affiliated to Savitribai Phule Pune University

- NAAC Accredited with A++ Grade
- NBA Accredited
- 'Best College Award 2019' by SPPU

Vision:

To be a globally renowned institution through excellence in engineering education for sustainable and holistic development

Our Mission is to:

M1: Empower students with cutting-edge technologies and global competencies

M2: Foster culture of research and entrepreneurial mindset

M3: Imbibe social and professional values

M4: Provide an inclusive environment for lifelong learning

Marathwada Mitramandal's

COLLEGE OF ENGINEERING

Karvenagar, Pune - 411052

Department of Computer Engineering

Vision

To excel in computer engineering education for sustainable development.

Mission

M1: To develop globally competent professionals through interdisciplinary learning and Center of Excellence.
M2: To empower research, innovation, and entrepreneurial thinking.
M3: To promote ethical values and holistic development.
M4: To prepare graduates for lifelong learning and dynamic careers.

Program Educational Objectives (PEOs)

PEO1: Graduates will demonstrate strong foundational knowledge to address global and multidisciplinary challenges.
PEO2: Graduates will engage in innovation, research, and entrepreneurship across multidisciplinary domains.
PEO3: Graduates will practice ethical conduct, professional and social responsibility.
PEO4: Graduates will exhibit lifelong learning and continuous skill enhancement.

Program Specific Outcomes (PSOs)

PSO1: Apply problem solving skills to design effective solutions in High Performance Computing, Artificial Intelligence and Cyber Security.
PSO2: Develop advanced skill-based solutions using standard Software Engineering practices.

Knowledge and Attitude Profile (WK)

WK1: A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.

WK2: Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.

WK3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

WK4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.

WK5: Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.

WK6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.

WK7: Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.

WK8: Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

WK9: Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

Program Outcomes (POs)

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.(WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.(WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.(WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8).

Rubrics for Lab Assessment (CAS)

Dimension	Scale				
	1	2	3	4	5
Regularity and punctuality	Did not Perform, submitted in time	Performed and submitted later than scheduled date	Performed on schedule; submitted two weeks late	Performed on schedule; submitted one week late	Performed and submitted as per schedule
Understanding and preparation for Objective	Neither shows any understanding of the objective nor can relate it to theory.	States the objective very vaguely	Can only state the objective but shows poor understanding	Understands objective but cannot place it in context of a theory	Understands objective and can relate it to an appropriate theory topic
Participation in performance and conduction of experiment	Does not participate in experiment	Performs the experiment only with the help from supervisor/others and is confused and untidy.	Performs the experiment with some supervisory help; but forgets some crucial reading and is confused and untidy.	Performs experiment on own without supervisory help; records all readings properly but untidy.	Performs experiment on his/her own without supervisory help; records all readings properly. Keeps the setup clean and tidy.
Post experiment skills	Cannot follow the procedure and do any work	Follows procedure half-heartedly	Follows right procedure; but cannot analyze data and interpret it	Follows right procedure and can analyze data and interpret it	Follows right procedure; can analyze data and interpret it with justification

Curriculum for Third Year of Computer Engineering (2019 Course), Savitribai Phule Pune University

Savitribai Phule Pune University
Third Year of Computer Engineering (2019 Course)
310256:Data Science and Big Data Analytics Laboratory

Home

Teaching Scheme	Credit:02	Examination Scheme and Marks
Practical: 04 Hours/Week		Term work: 50 Marks Practical: 25 Marks

Companion Course: Data Science and Big Data Analytics (310251)

Course Objectives:

- To understand principles of Data Science for the analysis of real time problems
- To develop in depth understanding and implementation of the key technologies in Data Science and Big Data Analytics
- To analyze and demonstrate knowledge of statistical data analysis techniques for decision-making
- To gain practical, hands-on experience with statistics programming languages and Big Data tools

Course Outcomes:

On completion of the course, learners will be able to

CO1: Apply principles of Data Science for the analysis of real time problems
CO2: Implement data representation using statistical methods
CO3: Implement and evaluate data analytics algorithms
CO4: Perform text preprocessing
CO5: Implement data visualization techniques
CO6: Use cutting edge tools and technologies to analyze Big Data

Guidelines for Instructor's Manual

The instructor's manual is to be developed as a reference and hands-on resource. It should include prologue (about University/program/ institute/ department/foreword/ preface), curriculum of the course, conduction and Assessment guidelines, topics under consideration, concept, objectives, outcomes, set of typical applications/assignments/ guidelines, and references.

Guidelines for Student's Laboratory Journal

The laboratory assignments are to be submitted by student in the form of journal. Journal consists of Certificate, table of contents, and handwritten write-up of each assignment (Title, Date of Completion, Objectives, Problem Statement, Software and Hardware requirements, Assessment grade/marks and assessor's sign, Theory- Concept in brief, algorithm, flowchart, test cases, Test Data Set(if applicable), mathematical model (if applicable), conclusion/analysis. Program codes with sample output of all performed assignments are to be submitted as softcopy. As a conscious effort and little contribution towards Green IT and environment awareness, attaching printed papers as part of write-ups and program listing to journal must be avoided. Use of DVD containing students programs maintained by Laboratory In-charge is highly encouraged. For reference one or two journals may be maintained with program prints in the Laboratory.

Guidelines for Laboratory /Term Work Assessment

Continuous assessment of laboratory work should be based on overall performance of Laboratory assignments by a student. Each Laboratory assignment assessment will assign grade/marks based on parameters, such as timely completion, performance, innovation, efficient codes, punctuality and

Guidelines for Practical Examination

Problem statements must be decided jointly by the internal examiner and external examiner. During practical assessment, maximum weightage should be given to satisfactory implementation of the problem statement. Relevant questions may be asked at the time of evaluation to test the student's understanding of the fundamentals, effective and efficient implementation. This will encourage, transparent evaluation and fair approach, and hence will not create any uncertainty or doubt in the minds of the students. So, adhering to these principles will consummate our team efforts to the promising start of student's academics.

Guidelines for Laboratory Conduction

http://collegecirculars.unipune.ac.in/sites/documents/Syllabus2021/Forms/AllItems.aspx #77/99

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The instructor is expected to frame the assignments by understanding the prerequisites, technological aspects, utility and recent trends related to the topic. The assignment framing policy need to address the average students and inclusive of an element to attract and promote the intelligent students. Use of open source software is encouraged. Based on the concepts learned. Instructor may also set one assignment or mini-project that is suitable to respective branch beyond the scope of syllabus.
Set of suggested assignment list is provided in groups- A and B. Each student must perform 13 assignments (10 from group A, 3 from group B), 2 mini project from Group C
Operating System recommended :- 64-bit Open source Linux or its derivative
Programming tools recommended: - JAVA/Python/R/Scala

Virtual Laboratory:

- "Welcome to Virtual Labs - A MHRD Govt of india Initiative"
- http://cse20-iiith.vlabs.ac.in/List%20of%20Experiments.html?domain=Computer%20Science

Suggested List of Laboratory Experiments/Assignments

Assignments from all Groups (A,B,C) are compulsory.

Sr. No.	Group A : Data Science
1.	**Data Wrangling, I** Perform the following operations using Python on any open source dataset (e.g., data.csv) 1. Import all the required Python Libraries. 2. Locate an open source data from the web (e.g. https://www.kaggle.com). Provide a clear description of the data and its source (i.e., URL of the web site). 3. Load the Dataset into pandas data frame. 4. Data Preprocessing: check for missing values in the data using pandas insult(), describe() function to get some initial statistics. Provide variable descriptions. Types of variables etc. Check the dimensions of the data frame. 5. Data Formatting and Data Normalization: Summarize the types of variables by checking the data types (i.e., character, numeric, integer, factor, and logical) of the variables in the data set. If variables are not in the correct data type, apply proper type conversions. 6. Turn categorical variables into quantitative variables in Python. In addition to the codes and outputs, explain every operation that you do in the above steps and explain everything that you do to import/read/scrape the data set.
2.	**Data Wrangling II** Create an "Academic performance" dataset of students and perform the following operations using Python. 1. Scan all variables for missing values and inconsistencies. If there are missing values and/or inconsistencies, use any of the suitable techniques to deal with them. 2. Scan all numeric variables for outliers. If there are outliers, use any of the suitable techniques to deal with them. 3. Apply data transformations on at least one of the variables. The purpose of this transformation should be one of the following reasons: to change the scale for better understanding of the variable, to convert a non-linear relation into a linear one, or to decrease the skewness and convert the distribution into a normal distribution. Reason and document your approach properly.

3.	**Descriptive Statistics - Measures of Central Tendency and variability** Perform the following operations on any open source dataset (e.g., data.csv) 1. Provide summary statistics (mean, median, minimum, maximum, standard deviation) for a dataset (age, income etc.) with numeric variables grouped by one of the qualitative (categorical) variable. For example, if your categorical variable is age groups and quantitative variable is income, then provide summary statistics of income grouped by the age groups. Create a list that contains a numeric value for each response to the categorical variable. 2. Write a Python program to display some basic statistical details like percentile, mean, standard deviation etc. of the species of 'Iris-setosa', 'Iris-versicolor' and 'Iris-versicolor' of iris.csv dataset. Provide the codes with outputs and explain everything that you do in this step.
4.	**Data Analytics I** Create a Linear Regression Model using Python/R to predict home prices using Boston Housing Dataset (https://www.kaggle.com/c/boston-housing). The Boston Housing dataset contains information about various houses in Boston through different parameters. There are 506 samples and 14 feature variables in this dataset. The objective is to predict the value of prices of the house using the given features.
5.	**Data Analytics II** 1. Implement logistic regression using Python/R to perform classification on Social_Network_Ads.csv dataset. 2. Compute Confusion matrix to find TP, FP, TN, FN, Accuracy, Error rate, Precision, Recall on the given dataset.
6.	**Data Analytics III** 1. Implement Simple Naïve Bayes classification algorithm using Python/R on iris.csv dataset. 2. Compute Confusion matrix to find TP, FP, TN, FN, Accuracy, Error rate, Precision, Recall on the given dataset.
7.	**Text Analytics** 1. Extract Sample document and apply following document preprocessing methods: Tokenization, POS Tagging, stop words removal, Stemming and Lemmatization. 2. Create representation of document by calculating Term Frequency and Inverse Document Frequency.
8.	**Data Visualization I** 1. Use the inbuilt dataset 'titanic'. The dataset contains 891 rows and contains information about the passengers who boarded the unfortunate Titanic ship. Use the Seaborn library to see if we can find any patterns in the data. 2. Write a code to check how the price of the ticket (column name: 'fare') for each passenger is distributed by plotting a histogram.
9.	**Data Visualization II** 1. Use the inbuilt dataset 'titanic' as used in the above problem. Plot a box plot for distribution of age with respect to each gender along with the information about whether they survived or not. (Column names : 'sex' and 'age') 2. Write observations on the inference from the above statistics.

10.	**Data Visualization III** Download the Iris flower dataset or any other dataset into a DataFrame. (e.g., https://archive.ics.uci.edu/ml/datasets/Iris). Scan the dataset and give the inference as: 1. List down the features and their types (e.g., numeric, nominal) available in the dataset. 2. Create a histogram for each feature in the dataset to illustrate the feature distributions. 3. Create a box plot for each feature in the dataset. 4. Compare distributions and identify outliers.
	Group B- Big Data Analytics – JAVA/SCALA (Any three)
1.	Write a code in JAVA for a simple Word Count application that counts the number of occurrences of each word in a given input set using the Hadoop Map-Reduce framework on local-standalone set-up.
2.	Design a distributed application using Map-Reduce which processes a log file of a system.
3.	Locate dataset (e.g., sample weather.txt) for working on weather data which reads the text input files and finds average for temperature, dew point and wind speed.
4.	Write a simple program in SCALA using Apache Spark framework
	Group C- Mini Projects/ Case Study – PYTHON/R (Any TWO Mini Project)
1.	Write a case study on Global Innovation Network and Analysis (GINA). Components of analytic plan are 1. Discovery business problem framed, 2. Data, 3. Model planning analytic technique and 4. Results and Key findings.
2.	Use the following dataset and classify tweets into positive and negative tweets. https://www.kaggle.com/ruchi798/data-science-tweets
3.	Develop a movie recommendation model using the scikit-learn library in python. Refer dataset https://github.com/rashida048/Some-NLP-Projects/blob/master/movie_dataset.csv
4.	Use the following covid vaccine statewise.csv dataset and perform following analytics on the given dataset https://www.kaggle.com/sudalairajkumar/covid19-in-india?select=covid_vaccine_statewise.csv a. Describe the dataset b. Number of persons state wise vaccinated for first dose in India c. Number of persons state wise vaccinated for second dose in India d. Number of Males vaccinated d. Number of females vaccinated
5.	Write a case study to process data driven for Digital Marketing **OR** Health care systems with Hadoop Ecosystem components as shown. (Mandatory) • HDFS: Hadoop Distributed File System • YARN: Yet Another Resource Negotiator • MapReduce: Programming based Data Processing • Spark: In-Memory data processing • PIG, HIVE: Query based processing of data services • HBase: NoSQL Database (Provides real-time reads and writes) • Mahout, Spark MLLib: (Provides analytical tools) Machine Learning algorithm libraries • Solar, Lucene: Searching and Indexing
	Learning Resources

CO PO and PSO Mapping

A. Course Outcome

CO. No.	Course Outcomes	Knowledge Levels
310256.1	Apply principles of Data Science for the analysis of real-time problems	**BTL 3**
310256.2	Apply statistical methods for data representation	**BTL 3**
310256.3	Evaluate data analytics algorithms	**BTL 5**
310256.4	Demonstrate text preprocessing	**BTL 3**
310256.5	Determine data visualization techniques	**BTL 3**
310256.6	Design Big Data Using cutting edge tools and technologies	**BTL 6**

B. CO-PO-PSO mapping

	PROGRAM OUTCOMES											PSO	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO1	1	2	2	1	-	-	-	-	-	-	-	1	-
CO2	2	2	-	1	-	-	-	-	-	-	-	1	-
CO3	2	1	2	1	-	2	-	-	-	-	1	1	-
CO4	1	1	1	2	2	-	-	-	-	-	-	1	-
CO5	1	1	1	1	2	-	-	-	-	-	1	1	-
CO6	1	1	1	2	2	-	-	-	-	-	1	1	-

INDEX				
Sr. No.		**List of Assignments**	**CO**	**PO**
Gr. A	1	Data Wrangling –Part -I	1	1,2,3,4
	2	Data Wrangling –Part -II	1	1,2,3,4
	3	Statistical Evaluation –Mean, Mode, Median Std. Dev. etc.	2	1,2,4
	4	Data Analytics- I - Linear Regression	3	1,2,3,4,6,11
	5	Data Analytics- II – Logistic Regression	3	1,2,3,4,6,11
	6	Data Analytics- III- Naïve Bayes	3	1,2,3,4,6,11
	7	Text analytics- POS tagging, stop Word Removal,stemming and lemmatization	4	1,2,3,4,5
	8	Data Visualization -I	5	1,2,3,4,5,11
	9	Data Visualization -II	5	1,2,3,4,5,11
	10	Data Visualization -III	5	1,2,3,4,5,11
Gr. B	1	Hadoop Framework for Word Count-standalone setup	6	1,2,3,4,5,11
	2	Design of Distributed Application -Processing Log file using Map-Reduce Framework	6	1,2,3,4,5,11
	3	Weather Data Forecasting –Map Reduce	6	1,2,3,4,5,11
Mini Project			1-6	1,2,3,4,5,6,11
VLAB			4	1,2,3,4,5

CERTIFICATE

Certified that Mr./Ms._____ of Class TE Computer Engineering Roll No._____ has completed the Journal assignments in the subject DSBDA Lab during the academic year 2023-24 Sem II.

Signature of the Faculty **Signature of Head of the Department**

Date:

Software Required:

1 64 bit open source operating system- Ubuntu
2 JAVA
3 PYTHON/R/SCALA

Write-ups must include:

- Group
- Assignment No.
- Aim
- Problem Statement
- Prerequisites
- Course Objectives
- Course Outcomes
- Theory(in brief)
- Test Cases
- Conclusion
- FAQs:
- Output: Soft copy of program with output.

GROUP: A

ASSIGNMENT NO: 1

PROBLEM STATEMENT: Perform the following operations using Python on any open source dataset (e.g., data.csv)

1. Import all the required Python Libraries.
2. Locate open source data from the web (e.g. https://www.kaggle.com). Provide a clear description of the data and its source (i.e., URL of the web site).
3. Load the Dataset into the pandas data frame.
4. Data Preprocessing: check for missing values in the data using pandas insult(), describe() function to get some initial statistics. Provide variable descriptions. Types of variables
etc. Check the dimensions of the data frame.
5. Data Formatting and Data Normalization: Summarize the types of variables by checking the data types (i.e., character, numeric, integer, factor, and logical) of the variables in the data set. If variables are not in the correct data type, apply proper type conversions.
6. Turn categorical variables into quantitative variables in Python.

In addition to the codes and outputs, explain every operation that you do in the above steps and explain everything that you do to import/read/scrape the data set.

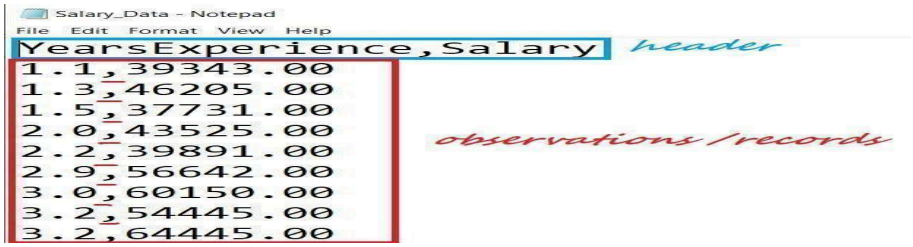
COURSE OBJECTIVES: To understand principles of Data Science for the analysis of real time problems

COURSE OUTCOMES: Apply principles of Data Science for the analysis of real time problems

THEORY:

1. What is a CSV?

CSV stands for "Comma Separated Values." It is the simplest form of storing data in tabular form as plain text. It is important to know to work with CSV
because we mostly rely on CSV data in our day-to-day lives as data scientists.

Structure of CSV: 

We have a file named "Salary_Data.csv." The first line of a CSV file is the header and contains the names of the fields/features.
After the header, each line of the file is an observation/a record. The values of a record are separated by "comma."

2. Reading a CSV

CSV files can be handled in multiple ways in Python.

2.1 Using csv.reader

Reading a CSV using Python's inbuilt module called **csv** using **csv.reader** object. Steps to read a CSV file:

1. Import the csv library

```
import csv
```

2. Open the CSV file

The .**open()** method in python is used to open files and return a file object.

```
file = open('Salary_Data.csv')

type(file)
```

The type of file is "**_io.TextIOWrapper**" which is a file object that is returned by the **open()** method.

3. Use the csv.reader object to read the CSV file

```
csvreader = csv.reader(file)
```

4. Extract the field names

Create an empty list called header. Use the next() method to obtain the header. The .next() method returns the current row and moves to the next row.

The first time you run next() it returns the header and the next time you run it returns the first record and so on.

```
header = []
header =
next(csvreader)
header
```

```
['YearsExperience', 'Salary']
```

5. Extract the rows/records

Create an empty list called rows and iterate through the csvreader object and append each row to the rows list.

```
rows = []
for row in
    csvreader:
    rows.append(r
    ow)
rows

data= pd.read_csv("Salary_Data.csv")

data
```

3. Extract the field names

.columns is used to obtain the header/field names. Data.columns

```
Index(['YearsExperience', 'Salary'], dtype='object')
```

	YearsExperience	Salary
0	1.1	39343.0
1	1.3	46205.0
2	1.5	37731.0
3	2.0	43525.0
4	2.2	39891.0
5	2.9	56642.0
6	3.0	60150.0
7	3.2	54445.0
8	3.2	64445.0
9	3.7	57189.0
10	3.9	63218.0
11	4.0	55794.0
12	4.0	56957.0
13	4.1	57081.0
14	4.5	61111.0
15	4.9	67938.0
16	5.1	66029.0
17	5.3	83088.0

4. Extract the rows

All the data of a data frame can be accessed using the field names.data.Salary

```
0      39343.0
1      46205.0
2      37731.0
3      43525.0
4      39891.0
5      56642.0
6      60150.0
7      54445.0
8      64445.0
9      57189.0
10     63218.0
11     55794.0
12     56957.0
13     57081.0
14     61111.0
15     67938.0
16     66029.0
17     83088.0
18     81363.0
19     93940.0
20     91738.0
21     98273.0
22    101302.0
23    113812.0
24    109431.0
25    105582.0
26    116969.0
27    112635.0
28    122391.0
29    121872.0
Name: Salary, dtype: float64
```

3. Writing to a CSV file

We can write to a CSV file in multiple ways.

3.1 Using csv.writer

Let's assume we are recording 3 Students data(Name, M1 Score, M2 Score)

header = ['Name', 'M1 Score', 'M2 Score']
data = [['Alex', 62, 80], ['Brad', 45, 56], ['Joey', 85, 98]]

Steps of writing to a CSV file:

1. Import csv library

import csv

2. Define a filename and Open the file using open()

3. Create a csvwriter object using csv.writer()

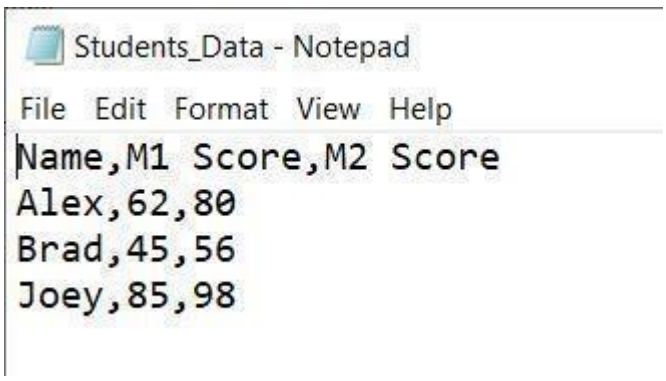
4. Write the header

5. Write the rest of the data

code for steps 2-5

```
filename = 'Students_Data.csv'
with open(filename, 'w', newline="") as file:
    csvwriter = csv.writer(file) # 2. create a csvwriter object
    csvwriter.writerow(header) # 4. write the header
    csvwriter.writerows(data) # 5. write the rest of the data
```

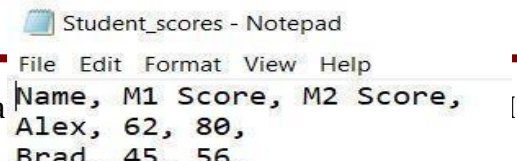
Below is how our CSV file looks.



3.2 Using .writelines()

Iterate through each list and convert the list elements to a string and write to the csv file.

```
header = ['Name', 'M1 Score', 'M2 Score']
data = [['Alex', 62, 80], ['Brad', 45, 56], ['Joey', 85, 98]]
filename = 'Student_scores.csv'
with open(filename, 'w') as file:
    for header in header:
        file.write(str(header)+', ')
    file.write('n')
    for row in data:
        for x in row:
            file.write(str(x)+', ')
        file.write('n')
```



3.3. Using pandas

Steps to writing to a CSV using pandas

1. Import pandas library

import pandas as pd

2. Create a pandas dataframe using pd.DataFrame **Syntax:** pd.DataFrame(data, columns)

The data parameter takes the records/observations and the columns parameter takes the columns/field names.

```
header = ['Name', 'M1 Score', 'M2 Score']
data = [['Alex', 62, 80], ['Brad', 45, 56], ['Joey', 85, 98]]
data = pd.DataFrame(data, columns=header)
```

3. Write to a CSV file using to_csv()

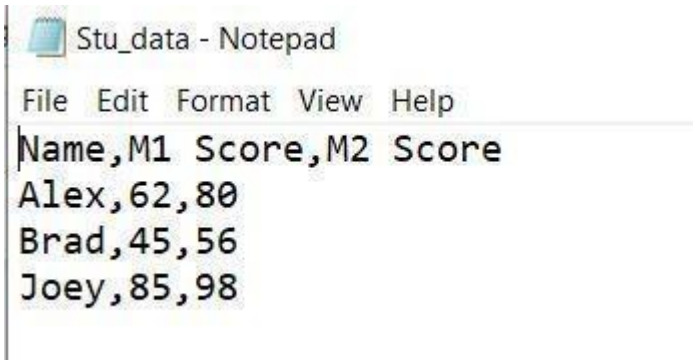
Syntax: DataFrame.to_csv(filename, sep=',', index=False)

**separator is ',' by default.

index=False to remove the index numbers.

data.to_csv('Stu_data.csv', index=False)

Below is how our CSV looks like



```
[['1.1', '39343.00'],
 ['1.3', '46205.00'],
 ['1.5', '37731.00'],
 ['2.0', '43525.00'],
 ['2.2', '39891.00'],
 ['2.9', '56642.00'],
 ['3.0', '60150.00'],
 ['3.2', '54445.00'],
 ['3.2', '64445.00'],
 ['3.7', '57189.00'],
 ['3.9', '63218.00'],
 ['4.0', '55794.00'],
 ['4.0', '56957.00'],
 ['4.1', '57081.00'],
 ['4.5', '61111.00'],
 ['4.9', '67938.00'],
 ['5.1', '66029.00'],
 ['5.3', '83088.00'],
 ['5.9', '81363.00'],
 ['6.0', '93940.00'],
 ['6.8', '91738.00'],
 ['7.1', '98273.00'],
 ['7.9', '101302.00'],
 ['8.2', '113812.00'],
 ['8.7', '109431.00'],
 ['9.0', '105582.00'],
 ['9.5', '116969.00'],
 ['9.6', '112635.00'],
 ['10.3', '122391.00'],
 ['10.5', '121872.00']]
```

6. Close the file

.close() method is used to close the opened file. Once it is closed, we cannot perform any operations on it.

```
file.close()
```

Complete Code:

```
import csv
file = open("Salary_Data.csv")
csvreader = csv.reader(file)
header = next(csvreader)
```

```
print(header)
rows = []
for row in csvreader:
    rows.append(row)
print(rows)
file.close()
```

Naturally, we might forget to close an open file. To avoid that we can use the **with()** statement to automatically release the resources. In simple terms, there is no need to call the .close() method if we are using the with() statement.

Implementing the above code using with() statement:
Syntax: with open(filename, mode) as alias_filename:

Modes:

‘r’ – to read an existing file,
‘w’ – to create a new file if the given file doesn’t exist and write to it,
‘a’ – to append to existing file content,
‘+’ – to create a new file for reading and writing

```
import csv
rows = []
with open("Salary_Data.csv", 'r) as file:
    csvreader = csv.reader(file)
    header = next(csvreader)
    for row in csvreader:
        rows.append(row)
print(header)
print(rows)
```

```
['YearsExperience', 'Salary']
[['1.1', '39343.00'], ['1.3', '46205.00'], ['1.5', '37731.00'], ['2.0', '43525.00'], ['2.2', '39891.00'], ['2.9', '56642.0
0'], ['3.0', '60150.00'], ['3.2', '54445.00'], ['3.2', '64445.00'], ['3.7', '57189.00'], ['3.9', '63218.00'], ['4.0', '5579
4.00'], ['4.0', '56957.00'], ['4.1', '57081.00'], ['4.5', '61111.00'], ['4.9', '67938.00'], ['5.1', '66029.00'], ['5.3', '83
088.00'], ['5.9', '81363.00'], ['6.0', '93940.00'], ['6.8', '91738.00'], ['7.1', '98273.00'], ['7.9', '101302.00'], ['8.2',
'113812.00'], ['8.7', '109431.00'], ['9.0', '105582.00'], ['9.5', '116969.00'], ['9.6', '112635.00'], ['10.3', '122391.00'],
['10.5', '121872.00']]
```

2.2 Using .readlines()
Now the question is – “Is it possible to fetch the header, rows using only open() and with() statements and without the csv library?” Let’s see…

.readlines() method is the answer. It returns all the lines in a file as a list. Each item of the list is a row of our CSV file.
The first row of the file.readlines() is the header and the rest of them are the records.

```
with open('Salary_Data.csv') as file:
    content = file.readlines()
header = content[:1]
rows = content[1:]
```

print(header)

print(rows)

```
['YearsExperience,Salary\n']
['1.1,39343.00\n', '1.3,46205.00\n', '1.5,37731.00\n', '2.0,43525.00\n', '2.2,39891.00\n', '2.9,56642.00\n', '3.0,60150.00
\n', '3.2,54445.00\n', '3.2,64445.00\n', '3.7,57189.00\n', '3.9,63218.00\n', '4.0,55794.00\n', '4.0,56957.00\n', '4.1,57081.
00\n', '4.5,61111.00\n', '4.9,67938.00\n', '5.1,66029.00\n', '5.3,83088.00\n', '5.9,81363.00\n', '6.0,93940.00\n', '6.8,9173
8.00\n', '7.1,98273.00\n', '7.9,101302.00\n', '8.2,113812.00\n', '8.7,109431.00\n', '9.0,105582.00\n', '9.5,116969.00\n',
'9.6,112635.00\n', '10.3,122391.00\n', '10.5,121872.00\n']
```

**The 'n' from the output can be removed using .strip() method.

What if we have a huge dataset with hundreds of features and thousands of records. Would it be possible to handle lists??

Here comes the pandas library into the picture.

2.3 Using pandas

Steps of reading CSV files using pandas

1. Import pandas library

import pandas as pd

2. Load CSV files to pandas using read_csv()

Basic Syntax: pandas.read_csv(filename, delimiter=',')

data= pd.read_csv("Salary_Data.csv")

data

	YearsExperience	Salary
0	1.1	39343.0
1	1.3	46205.0
2	1.5	37731.0
3	2.0	43525.0
4	2.2	39891.0
5	2.9	56642.0
6	3.0	60150.0
7	3.2	54445.0
8	3.2	64445.0
9	3.7	57189.0
10	3.9	63218.0
11	4.0	55794.0
12	4.0	56957.0
13	4.1	57081.0
14	4.5	61111.0
15	4.9	67938.0
16	5.1	66029.0
17	5.3	83088.0

OUTPUT:

CONCLUSION: Students will be able to apply mathematics, data science fundamentals to solve real time problems

FAQs:

1. Enlist the standard methods of data collection.
2. What is the need of data preprocessing?
3. Compare the performance of data analysis a)Random data collected
b) Preprocessed data
4. What are the methods of data reduction?
5. What are the ways to check duplicates in data.

Group A
Assignment No. 2

TITLE: Data Wrangling Part- II

PROBLEM STATEMENT: Create an "Academic performance" dataset of students and perform the following operations using Python.
1. Scan all variables for missing values and inconsistencies. If there are missing values and/or inconsistencies, use any of the suitable techniques to deal with them.
2. Scan all numeric variables for outliers. If there are outliers, use any of the suitable techniques to deal with them.
3. Apply data transformations on at least one of the variables. The purpose of this transformation should be one of the following reasons: to change the scale for better understanding of the variable, to convert a non-linear relation into a linear one, or to decrease the skewness and convert the distribution into a normal distribution. Reason and document your approach properly.

COURSE OBJECTIVE: To understand principles of Data Science for the analysis of real time problems

COURSE OUTCOME: :Students will be able to analyse and identify different types of statistical methods for data analysis

THEORY:

Working with Missing Data in Pandas

Missing Data can occur when no information is provided for one or more items or for a whole unit. Missing Data is a very big problem in real-life scenarios. Missing Data can also refer to as NA(Not Available) values in pandas. In DataFrame sometimes many datasets simply arrive with missing data, either because it exists and was not collected or it never existed. For Example, Suppose different users being surveyed may choose not to share their income, some users may choose not to share the address in this way many datasets went missing.

In Pandas missing data is represented by two value:

- None: None is a Python singleton object that is often used for missing data in Python code.
- NaN : NaN (an acronym for Not a Number), is a special floating-point value recognized by all systems that use the standard IEEE floating-point representation

Pandas treat None and NaN as essentially interchangeable for indicating missing or null values. To facilitate this convention, there are several useful functions for detecting, removing, and replacing null values in Pandas DataFrame :

- isnull()
- notnull()
- dropna()
- fillna()
- replace()
- interpolate()

Checking for missing values using isnull() and notnull()
In order to check missing values in Pandas DataFrame, we use a function isnull() and notnull(). Both functions help in checking whether a value is NaN or not. These functions can also be used in Pandas Series in order to find null values in a series.

Filling missing values using fillna(), replace()

In order to fill null values in a datasets, we use fillna(), replace() and interpolate() functions. These functions replace NaN values with some value of their own. All these functions help in filling null values in datasets of a DataFrame. Interpolate() function is basically used to fill NA values in the dataframe but it uses various interpolation techniques to fill the missing values rather than hard-coding the value.
Code #1: Filling null values with a single value

```
# importing pandas package

import pandas as pd

   # making data frame from csv file

data = pd.read_csv("employees.csv")

# Printing the first 10 to 24 rows of

# the data frame for visualization

    data[10:25]
```

10	Louise	Female	8/12/1980	9:01 AM	63241	15.132	True	NaN
11	Julie	Female	10/26/1997	3:19 PM	102508	12.637	True	Legal
12	Brandon	Male	12/1/1980	1:08 AM	112807	17.492	True	Human Resources
13	Gary	Male	1/27/2008	11:40 PM	109831	5.831	False	Sales
14	Kimberly	Female	1/14/1999	7:13 AM	41426	14.543	True	Finance
15	Lillian	Female	6/5/2016	6:09 AM	59414	1.256	False	Product
16	Jeremy	Male	9/21/2010	5:56 AM	90370	7.369	False	Human Resources
17	Shawn	Male	12/7/1986	7:45 PM	111737	6.414	False	Product
18	Diana	Female	10/23/1981	10:27 AM	132940	19.082	False	Client Services
19	Donna	Female	7/22/2010	3:48 AM	81014	1.894	False	Product
20	Lois	No Gender	4/22/1995	7:18 PM	64714	4.934	True	Legal
21	Matthew	Male	9/5/1995	2:12 AM	100612	13.645	False	Marketing
22	Joshua	No Gender	3/8/2012	1:58 AM	90816	18.816	True	Client Services
23	NaN	Male	6/14/2012	4:19 PM	125792	5.042	NaN	NaN
24	John	Male	7/1/1992	10:08 PM	97950	13.873	False	Client Services

	First Name	Gender	Start Date	Last Login Time	Salary	Bonus %	Senior Management	Team
10	Louise	Female	8/12/1980	9:01 AM	63241	15.132	True	NaN
11	Julie	Female	10/26/1997	3:19 PM	102508	12.637	True	Legal
12	Brandon	Male	12/1/1980	1:08 AM	112807	17.492	True	Human Resources
13	Gary	Male	1/27/2008	11:40 PM	109831	5.831	False	Sales
14	Kimberly	Female	1/14/1999	7:13 AM	41426	14.543	True	Finance
15	Lillian	Female	6/5/2016	6:09 AM	59414	1.256	False	Product
16	Jeremy	Male	9/21/2010	5:56 AM	90370	7.369	False	Human Resources
17	Shawn	Male	12/7/1986	7:45 PM	111737	6.414	False	Product
18	Diana	Female	10/23/1981	10:27 AM	132940	19.082	False	Client Services
19	Donna	Female	7/22/2010	3:48 AM	81014	1.894	False	Product
20	Lois	NaN	4/22/1995	7:18 PM	64714	4.934	True	Legal
21	Matthew	Male	9/5/1995	2:12 AM	100612	13.645	False	Marketing
22	Joshua	NaN	3/8/2012	1:58 AM	90816	18.816	True	Client Services
23	NaN	Male	6/14/2012	4:19 PM	125792	5.042	NaN	NaN

First step of cleaning the data is to scan all variables for missing values and inconsistencies. If there are missing values and/or inconsistencies, usage of any of the suitable techniques is done to deal with them.Secondly, scan all numeric variables for outliers. If there are outliers, usage of any of the

suitable techniques is done to deal with them. After that, apply data transformations on at least one of the variables. The purpose of this transformation should be one of the following reasons: to change the scale for better understanding of the variable, to convert a non-linear relation into a linear one, or to decrease the skewness and convert the distribution into a normal distribution

FAQs:

1. Enlist the standard methods of data collection.
2. What is the need of data preprocessing?
3. Compare the performance of data analysis a)Random data collected
b) Preprocessed data
4. What are the methods of data reduction.
5. What are the ways to check duplicates in data.

.

Group A
Assignment No. 3

TITLE: Statistical Evaluation –Mean, Mode, Median Std. Dev. etc

PROBLEM STATEMENT: Perform the following operations on any open source dataset (e.g., data.csv)

1. Provide summary statistics (mean, median, minimum, maximum, standard deviation) for a dataset (age, income etc.) with numeric variables grouped by one of the qualitative (categorical) variable. For example, if your categorical variable is age groups and quantitative variable is income, then provide summary statistics of income grouped by the age groups. Create a list that contains a numeric value for each response to the categorical variable.
2. Write a Python program to display some basic statistical details like percentile, mean, standard deviation etc. of the species of 'Iris-setosa', 'Iris-versicolor' and 'Iris-versicolor' of iris.csv dataset.

COURSE OBJECTIVE: To understand principles of Data Science for the analysis of real time problems

COURSE OUTCOME: :Students will be able to analyse and identify different types of statistical methods for data analysis.

THEORY:Measure of Central Tendency:
Mean:

- simplest , widely used measure.
- summarising the essential features of a series and in enabling data to be compared.
- relatively stable measure of central tendency.
- The mean of the population is denoted as μ (myoo) and x-bar for mean of sample

$$\text{Mean (or } \overline{X})^* = \frac{\sum X_i}{n} = \frac{X_1 + X_2 + \ldots + X_n}{n}$$

where $\overline{X}$ = The symbol we use for mean (pronounced as X bar)
$\sum$ = Symbol for summation
X_i = Value of the ith item X, $i = 1, 2, \ldots, n$
n = total number of items

Example:

$$\bar{x} = \frac{30+36+47+50+52+52+56+60+63+70+70+110}{12}$$

$$= \frac{696}{12} = 58.$$

Median:

- Median is the value of the middle item of a series when it is arranged in ascending or descending order of magnitude.
- It divides the series into two halves
- In one half all items are less than median,
- In the other half all items have values higher than median.
- It is exactly 50% of the individuals in the distribution who score at or below the median.
- It is equivalent to the 50th percentile
- Eg. –
- 3,5,8,10,11 median is 8 (odd)
- 3,3,4,5,7,8 median is (4+5)/2 = 4.5
- The median is expensive to compute if there is too large data.
- can approximate the median of the entire data set (e.g., the median salary) by interpolation.

Mode:

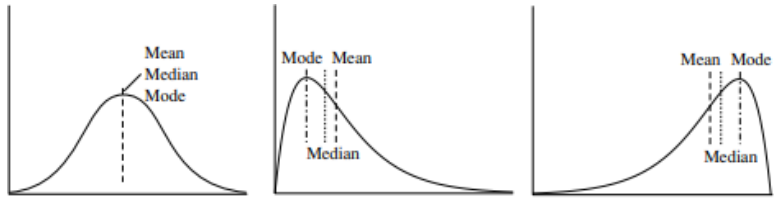
- The mode in a distribution is that item around which there is maximum concentration.
- Mode is the size of the item which has the maximum frequency,
- Mode is a positional average and is not affected by the values of extreme items.
- useful in all situations where to eliminate the effect of extreme variations.
- For example,
- a manufacturer of shoes interested in finding out the size most in demand.
- It is possible for the greatest frequency to correspond to several different values, which results in more than one mode.
- Data sets with one, two, or three modes are respectively called unimodal, bimodal, and trimodal.
- In general, a data set with two or more modes is multimodal.
- At the other extreme, if each data value occurs only once, then there is no mode.

e. g. –2,3,5,6,5,7,5
From the given dataset, 5 has the highest frequency and hence, the mode is 5
5,9,1,3,7,6,8
From the given dataset, every data is occurring once and hence, there is no repetition. Therefore, the dataset doesn't have a mode.

Midrange:

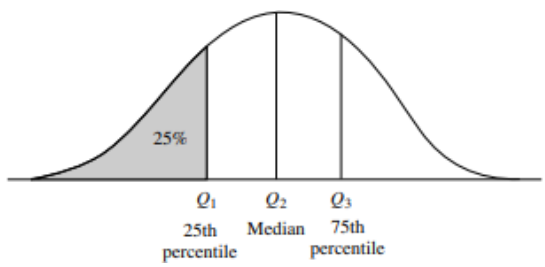
- The midrange is the average of the largest and smallest values in the set.
- Perfect symmetric data distribution, the mean, median, and mode are all at the same center value.
- Data in most real applications are not symmetric.
- Positively skewed, where the mode occurs at a value that is smaller than the median,

- Negatively skewed, where the mode occurs at a value greater than the median.



(a) Symmetric data (b) Positively skewed data (c) Negatively skewed data

Range, Quartiles, and Interquartile Range:

- Let x1,x2,...,xN be a set of observations for some numeric attribute, X. The range of the set is the difference between the largest (max()) and smallest (min()) values.
- Suppose that the data for attribute X are sorted in increasing numeric order. Imagine that we can pick certain data points so as to split the data distribution into equal-size consecutive sets. These data points are called quantiles.
- Quantiles are points taken at regular intervals of a data distribution, dividing it into essentially equal size consecutive sets.
- 

- The 2-quantile is the data point dividing the lower and upper halves of the data distribution. It corresponds to the median.
- The 4-quantiles are the three data points that split the data distribution into four equal parts; each part represents one-fourth of the data distribution. They are more commonly referred to as quartiles.
- The median, quartiles, and percentiles are the most widely used forms of quantiles.
- The quartiles give an indication of a distribution's center, spread, and shape. The first quartile, denoted by Q1, is the 25th percentile.
- It cuts off the lowest 25% of the data.
- The third quartile, denoted by Q3, is the 75th percentile—it cuts off the lowest 75% (or highest 25%) of the data.
- The second quartile is the 50th percentile.
- As the median, it gives the center of the data distribution.
- simple measure of spread : the interquartile range (IQR)
- `IQR = Q3 – Q1.`

Mean Deviation:

- Mean deviation is the average of difference of the values of items from some average of the series.

$$\text{Mean deviation from mean } (\delta_{\bar{X}}) = \frac{\Sigma|X_i - \bar{X}|}{n}, \quad \text{if deviations, } |X_i - \bar{X}|, \text{ are obtained from arithmetic average.}$$

or

$$\text{Mean deviation from median } (\delta_m) = \frac{\Sigma|X_i - M|}{n}, \quad \text{if deviations, } |X_i - M|, \text{ are obtained from median}$$

or

$$\text{Mean deviation from mode } (\delta_z) = \frac{\Sigma|X_i - Z|}{n}, \quad \text{if deviations, } |X_i - Z|, \text{ are obtained from mode.}$$

where δ = Symbol for mean deviation (pronounced as delta);
X_i = ith values of the variable X;
n = number of items;
$\bar{X}$ = Arithmetic average;
M = Median;
Z = Mode.

Variance and Standard Deviation:

- ***Standard deviation*** is the most widely used measure of dispersion of a series , denoted by ***the symbol 'σ'*** (pronounced as sigma).
- When we divide the standard deviation by the arithmetic average of the series, the resulting quantity is known as coefficient of standard deviation
- The standard deviation (along with several related measures like variance, coefficient of variation, etc.) is used mostly in research studies and is regarded as a very satisfactory measure of dispersion in a series.
- It is less affected by fluctuations of sampling.

$$\text{Standard deviation}^* (\sigma) = \sqrt{\frac{\Sigma(X_i - \bar{X})^2}{n}}$$

- A low standard deviation means that the data observations tend to be very close to the mean, while a high standard deviation indicates that the data are spread out over a large range of values.
- The variance of N observations, x1,x2,...,xN , for a numeric attribute X is

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^2 = \left(\frac{1}{N}\sum_{i=1}^{N}x_i^2\right) - \bar{x}^2,$$

where $\bar{x}$ is the mean value of the observations. The standard deviation, σ, of the observations is the square root of the variance, σ 2.

OUTPUT:

CONCLUSION: Students will design and develop big data solutions to solve complex and real time problems.

FAQs:

1. What are the central measures of data dispersion?
2. What is mean , mode, median?
3. What is std. deviation?
4. What is std. error?
5. Why measures of data dispersion are essential for data analysis.

Group A
Assignment No. 4

TITLE: Data Analytics- I - Linear Regression

PROBLEM STATEMENT: Create a Linear Regression Model using Python/R to predict home prices using Boston Housing .Dataset (https://www.kaggle.com/c/boston-housing). The Boston Housing dataset contains information about various houses in Boston through different parameters. There are 506 samples and 14 feature variables in this dataset.The objective is to predict the value of prices of the house using the given features.

COURSE OBJECTIVES: To understand principles of Data Science for the analysis of real time problems

COURSE OUTCOMES: Implement and evaluate data analytics algorithms

THEORY:

First, we will import the required libraries.Next, we will load the housing data from the scikit-learn library and understand it.

We print the value of the boston_dataset to understand what it contains. print(boston_dataset.keys()) gives.

dict_keys(['data', 'target', 'feature_names', 'DESCR'])

- data: contains the information for various houses
- target: prices of the house
- feature_names: names of the features
- DESCR: describes the dataset

To know more about the features use boston_dataset.DESCR The description of all the features is given below:

CRIM: Per capita crime rate by town
ZN: Proportion of residential land zoned for lots over 25,000 sq. ft
INDUS: Proportion of non-retail business acres per town
CHAS: Charles River dummy variable (= 1 if tract bounds river; 0 otherwise)
NOX: Nitric oxide concentration (parts per 10 million)
RM: Average number of rooms per dwelling
AGE: Proportion of owner-occupied units built prior to 1940
DIS: Weighted distances to five Boston employment centers
RAD: Index of accessibility to radial highways
TAX: Full-value property tax rate per $10,000
PTRATIO: Pupil-teacher ratio by town
B: $1000(Bk — 0.63)^2$, where Bk is the proportion of [people of African American descent] by town

LSTAT: Percentage of lower status of the population
MEDV: Median value of owner-occupied homes in $1000s

The prices of the house indicated by the variable MEDV is our target variable and the remaining are the feature variables based on which we will predict the value of a house.

We will now load the data into a pandas dataframe using pd.DataFrame. We then print the first 5 rows of the data using head()

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B	LSTAT
0	0.00632	18.0	2.31	0.0	0.538	6.575	65.2	4.0900	1.0	296.0	15.3	396.90	4.98
1	0.02731	0.0	7.07	0.0	0.469	6.421	78.9	4.9671	2.0	242.0	17.8	396.90	9.14
2	0.02729	0.0	7.07	0.0	0.469	7.185	61.1	4.9671	2.0	242.0	17.8	392.83	4.03
3	0.03237	0.0	2.18	0.0	0.458	6.998	45.8	6.0622	3.0	222.0	18.7	394.63	2.94
4	0.06905	0.0	2.18	0.0	0.458	7.147	54.2	6.0622	3.0	222.0	18.7	396.90	5.33

We can see that the target value MEDV is missing from the data. We create a new column of target values and add it to the dataframe.

Data preprocessing
After loading the data, it's a good practice to see if there are any missing values in the data. We count the number of missing values for each feature using isnull()

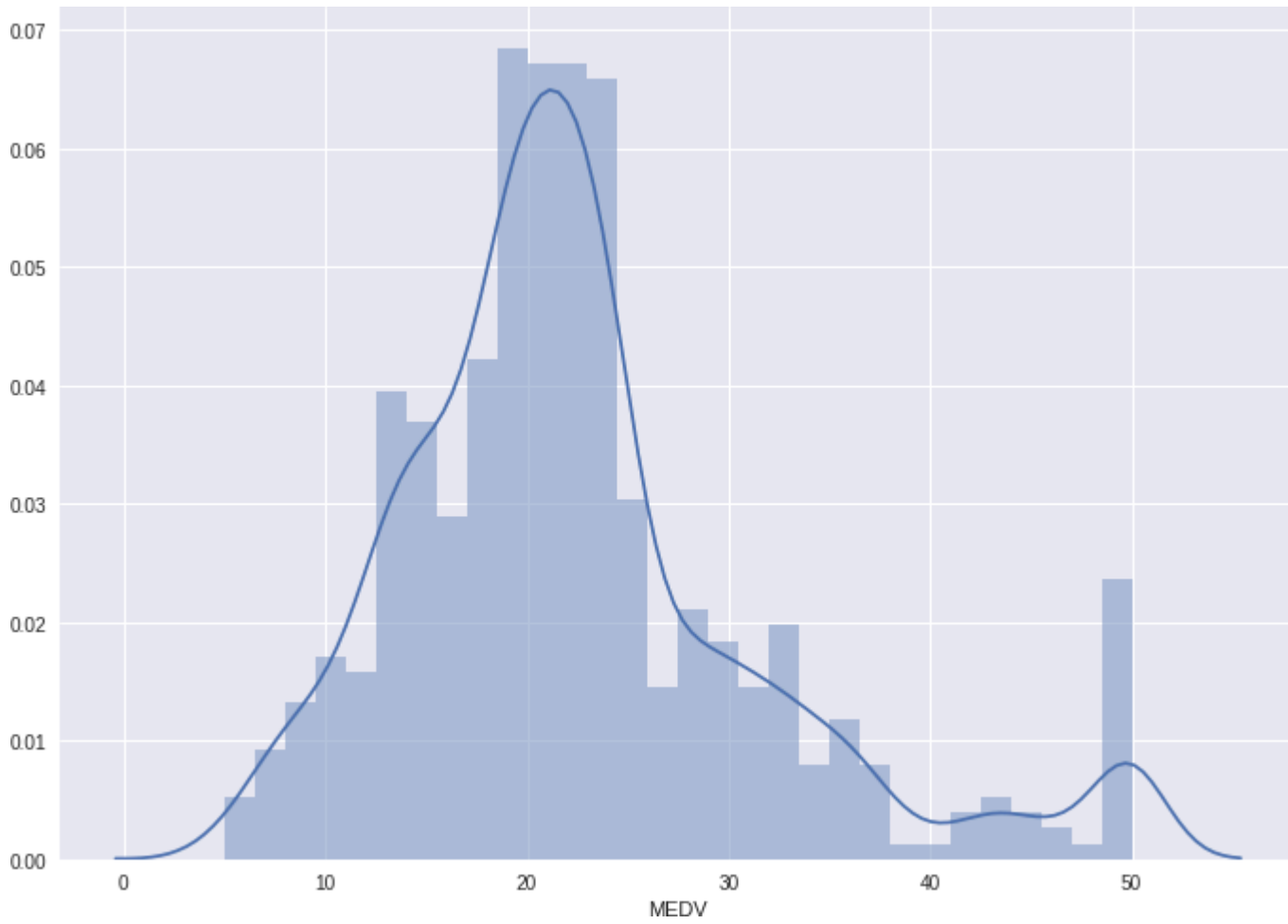
However, there are no missing values in this dataset as shown below.

```
CRIM       0
ZN         0
INDUS      0
CHAS       0
NOX        0
RM         0
AGE        0
DIS        0
RAD        0
TAX        0
PTRATIO    0
B          0
LSTAT      0
MEDV       0
dtype: int64
```

Exploratory Data Analysis

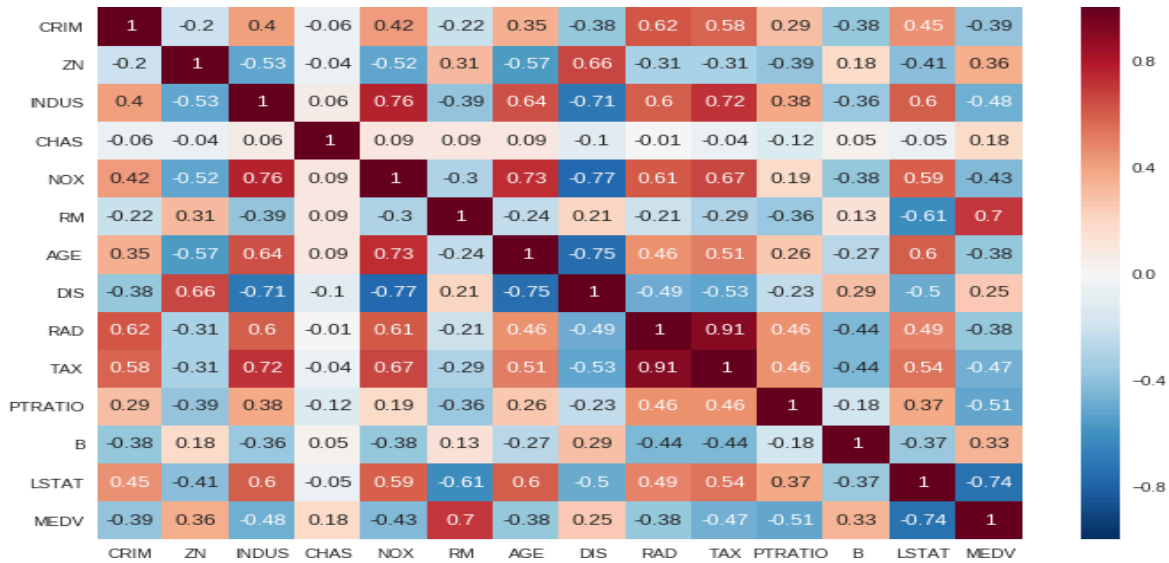
Exploratory Data Analysis is a very important step before training the model. In this section, we will use some visualizations to understand the relationship of the target variable with other features.

Let's first plot the distribution of the target variable MEDV. We will use the distplot function from the seaborn library.



We see that the values of MEDV are distributed normally with few outliers.

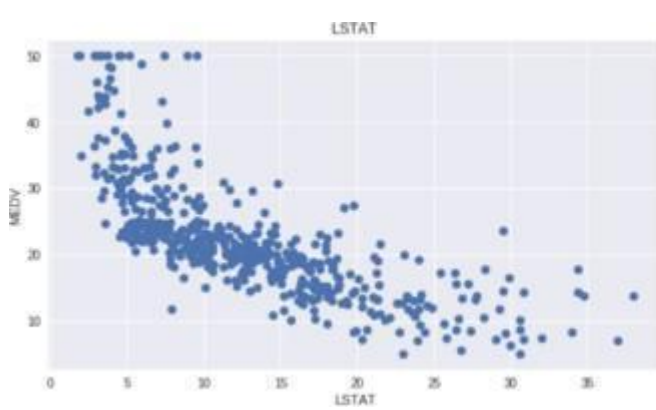
Next, we create a correlation matrix that measures the linear relationships between the variables. The correlation matrix can be formed by using the corr function from the pandas dataframe library. We will use the heatmap function from the seaborn library to plot the correlation matrix.

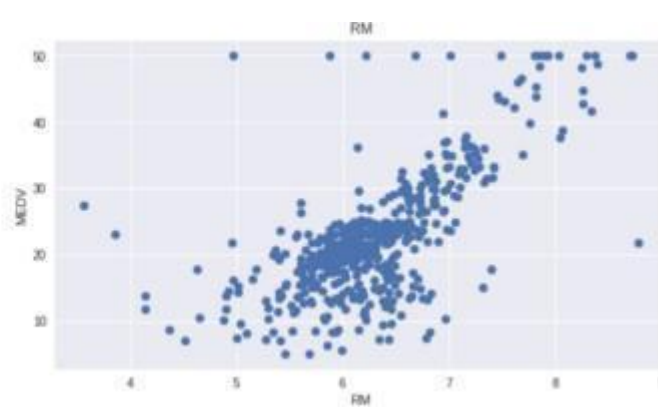


Observations:

- To fit a linear regression model, we select those features which have a high correlation with our target variable MEDV. By looking at the correlation matrix we can see that RM has a strong positive correlation with MEDV (0.7) whereas LSTAT has a high negative correlation with MEDV(-0.74).
- An important point in selecting features for a linear regression model is to check for multi-co-linearity. The features RAD, TAX have a correlation of 0.91. These feature pairs are strongly correlated to each other. We should not select both these features together for training the model. Check this for an explanation. Same goes for the features DIS and AGE which have a correlation of -0.75.

Based on the above observations we will RM and LSTAT as our features. Using a scatter plot let's see how these features vary with MEDV.





Observations:

- The prices increase as the value of RM increases linearly. There are few outliers and the data seems to be capped at 50.
- The prices tend to decrease with an increase in LSTAT. Though it doesn't look to be following exactly a linear line.

Preparing the data for training the model
We concatenate the LSTAT and RM columns using np.c_ provided by the numpy library.

Splitting the data into training and testing sets
Next, we split the data into training and testing sets. We train the model with 80% of the samples and test with the remaining 20%. We do this to assess the model's performance on unseen data. To split the data we use train_test_split function provided by scikit-learn library. We finally print the sizes of our training and test set to verify if the splitting has occurred properly.

(404, 2)
(102, 2)
(404,)
(102,)

Training and testing the model
We use scikit-learn's LinearRegression to train our model on both the training and test sets.

Model evaluation
We will evaluate our model using RMSE and R2-score.

The model performance for training set

RMSE is 5.6371293350711955
R2 score is 0.6300745149331701 **The model performance for testing set**

RMSE is 5.137400784702911
R2 score is 0.6628996975186952

This is good to start with. In the upcoming blogs, we will look at ways to increase the model's performance.

The complete Jupyter Notebook can be found here.

Here are a few places you can look for data

- https://www.kaggle.com/datasets
- https://toolbox.google.com/datasetsearch
- https://archive.ics.uci.edu/ml/datasets.html

OUTPUT:

CONCLUSION: Students will be able to use knowledge of software to investigate data analytics algorithms.

FAQs:

1. What is the use of data regression?
2. What is the special characteristic of linear regression?
3. Explain with an example an application of linear regression.
4. Which statistical knowledge you will use to apply linear regression to the given problem.
5. What is RMSE?

Group A
Assignment No. 5

TITLE: Data Analytics- II – Logistic Regression

PROBLEM STATEMENT: 1. Implement logistic regression using Python/R to perform classification on Social_Network_Ads.csv dataset.
2. Compute Confusion matrix to find TP, FP, TN, FN, Accuracy, Error rate, Precision, Recall on the given dataset.

COURSE OBJECTIVES: To understand principles of Data Science for the analysis of real time problems

COURSE OUTCOMES: Implement and evaluate data analytics algorithms

THEORY:

Logistic Regression: Classification techniques are an essential part of machine learning and data mining applications. Approximately 70% of problems in Data Science are classification problems. There are lots of classification problems that are available, but logistic regression is common and is a useful regression method for solving the binary classification problem. Another category of classification is Multinomial classification, which handles the issues where multiple classes are present in the target variable. For example, the IRIS dataset is a very famous example of multi-class classification. Other examples are classifying article/blog/document categories.

Logistic Regression can be used for various classification problems such as spam detection. Diabetes prediction, if a given customer will purchase a particular product or will they churn another competitor, whether the user will click on a given advertisement link or not, and many more examples are in the bucket.

For example, it can be used for cancer detection problems. It computes the probability of an event occurring. It is a special case of linear regression where the target variable is categorical in nature. It uses a log of odds as the dependent variable. Logistic Regression predicts the probability of occurrence of a binary event utilising a logit function.

1. Linear Regression Equation:

Where, y is a dependent variable and x1, x2 ... and Xn are explanatory variables.Sigmoid Function: Apply Sigmoid function on linear regression:

2. Differentiate between Linear and Logistic Regression

Linear regression gives you a continuous output, but logistic regression provides a constant
output. Linear regression is estimated using Ordinary Least Squares (OLS) while logistic regression is estimated using Maximum Likelihood Estimation (MLE) approach.

3. Sigmoid Function

The sigmoid function, also called logistic function, gives an 'S' shaped curve that can take any real-valued number and map it into a value between 0 and 1. If the curve goes to positive infinity,y predicted will become 1, and if the curve goes to negative infinity, y predicted will become 0.If the output of the sigmoid function is more than 0.5, we can classify the outcome as 1 or YES,
and if it is less than 0.5, we can classify it as 0 or NO. The outputcannotFor example: If the output is 0.75, we can say in terms of probability as: There is a 75 percent chance

that a patient will suffer from cancer.

4. Types of LogisticRegression

Binary Logistic Regression: The target variable has only two possible outcomes such as Spam or Not Spam, Cancer or No Cancer.

Multinomial Logistic Regression: The target variable has three or more nominal categories such as predicting the type of Wine.

Ordinal Logistic Regression: the target variable has three or more ordinal categories such as restaurant or product rating from 1 to 5.

5. Confusion Matrix Evaluation Metrics

Contingency table or Confusion matrix is often used to measure the performance of classifiers. A confusion matrix contains information about actual and predicted classifications done by a classification system. Performance of such systems is commonly evaluated using the data in the matrix.

Some Important measures derived from confusion matrix are:

- Number of positive (Pos) : Total number instances which are labelled as positive in a given dataset.
- Number of negative (Neg) : Total number instances which are labelled as negative in a given dataset.
- Number of True Positive (TP) : Number of instances which are actually labelled as positive and the predicted class by classifier is also positive.
- Number of True Negative (TN) : Number of instances which are actually labelled as negative and the predicted class by classifier is also negative.
- Number of False Positive (FP) : Number of instances which are actually labelled as negative and the predicted class by classifier is positive.
- Number of False Negative (FN): Number of instances which are actually labelled as positive and the class predicted by the classifier is negative.

TP

TN

- Accuracy: Accuracy is calculated as the number of correctly classified instances divided by total number of instances.

The ideal value of accuracy is 1, and the worst is 0. It is also calculated as the sum of true positive and true negative (TP + TN) divided by the total number of instances.

$acc = \frac{TP+TN}{TP+FP+TN+FN} = \frac{TP+TN}{Pos+Neg}$

- Error Rate: Error Rate is calculated as the number of incorrectly classified instances divided by total number of instances.

The ideal value of accuracy is 0, and the worst is 1. It is also calculated as the sum of false positive and false negative (FP + FN) divided by the total number of instances.

$err = \frac{FP+FN}{TP+FP+TN+FN} = \frac{FP+FN}{Pos+Neg}$ Or

$err = 1 - acc$

- Precision: It is calculated as the number of correctly classified positive instances divided by the total number of instances which are predicted positive. It is also called confidence value.

The ideal value is 1, whereas the worst is 0.

precision = TP
P+F

● Recall: .It is calculated as the number of correctly classified positive instances divided by the total number of positive instances. It is also called recall or sensitivity. The ideal value of sensitivity is 1, whereas the worst is 0.

It is calculated as the number of correctly classified positive instances divided by the total number
of positive instances.

recall = TP
P+F

Algorithm (Social Network Ads Dataset):

Step 1: Import libraries and create alias for Pandas, Numpy and Matplotlib

Step 2: Import the Social_Media_Adv Dataset

Step 3: Initialize the data frame

Step 4: Perform Data Preprocessing

● Convert Categorical to Numerical Values if applicable
● Check for Null Value
● Covariance Matrix to select the most promising features
● Divide the dataset into
Independent(X) and Dependent(Y)variables.
● Split the dataset into training and testing datasets
● Scale the Features if necessary.

Step 5: Use Logistic regression(Train the Machine) to Create Model

```
# import the class
from sklearn.linear_model import LogisticRegression
# instantiate the model (using the default parameters)
logreg = LogisticRegression()
# fit the model with data
logreg.fit(xtrain,ytrain)
# y_pred=logreg.predict(xtest)
```

Step 6: Predict the y_pred for all values of train_x and test_x

Step 7:Evaluate the performance of Model for train_y and test_y

Step 8: Calculate the required evaluation parameters

```
from sklearn.metrics import
precision_score,confusion_matrix,accuracy_score,recall_score
cm= confusion_matrix(ytest, y_pred)
```

Conclusion:

In this way we have done data analysis using logistic regression for Social Media Adv. and
evaluate the performance of model.

Value Addition: Visualising Confusion Matrix using Heatmap.

OUTPUT:

CONCLUSION: Students will design and develop different data nalytics algorithms to solve complex problems.

FAQs:

1) What is use of data regression?
2) What is the special characteristic of logistic regression?
3) Explain with an example an application of logistic regression.
4) Which statistical knowledge you will use to apply logistic regression to the given problem.
5) What is RMSE?

Group A
Assignment No. 6

TITLE: NAÏVE BAYES classification for social data .

PROBLEM STATEMENT:

1. Implement Simple Naïve Bayes classification algorithm using Python/R on iris.csv dataset.
2. Compute Confusion matrix to find TP, FP, TN, FN, Accuracy, Error rate, Precision, Recall on the given dataset.

COURSE OBJECTIVES:

To develop in depth understanding and implementation of the key technologies in Data Science and Big Data Analytics

COURSE OUTCOMES:

Implement and evaluate data analytics algorithms

THEORY:

Overview of Naive Bayes Classification

Naive Bayes is one such algorithm in classification that can never be overlooked upon due to its special characteristic of being "***naive***". It makes the assumption that features of a measurement are independent of each other.

For example, an animal may be considered as a cat if it has cat eyes, whiskers and a long tail. Even if these features depend on each other or upon the existence of the other features, all of these properties independently contribute to the probability that this animal is a cat and that is why it is known as 'Naive'.

According to Bayes Theorem, the various features are mutually independent. For two independent events, P(A,B) = P(A)P(B). This assumption of Bayes Theorem is probably never encountered in practice, hence it accounts for the "naive" part in Naive Bayes. Bayes' Theorem is stated as: ***P(a|b) = (P(b|a) * P(a)) / P(b).*** Where P(a|b) is the probability of a given b.

Let us understand this algorithm with a simple example. The Student will be a pass if he wears a "red" color dress on the exam day. We can solve it using the above discussed method of posterior probability.

By Bayes Theorem, *P(Pass| Red) = P(Red| Pass) * P(Pass) / P (Red).*

From the values, let us assume P (Red|Pass) = 3/9 = 0.33, P(Red) = 5/14 = 0.36, P(Pass)= 9/14 = 0.64.
Now, P (Pass| Red) = 0.33 * 0.64 / 0.36 = 0.60, which has higher probability.

In this way, Naive Bayes uses a similar method to predict the probability of different class based on various

attributes.

Problem Analysis

To implement the Naive Bayes Classification, we shall use a very famous Iris Flower Dataset that consists of 3 classes of flowers. In this, there are 4 independent variables namely the, ***sepal_length***, ***sepal_width***, ***petal_length*** and ***petal_width***. The dependent variable is the ***species*** which we will predict using the four independent features of the flowers.

There are 3 classes of species namely *setosa, versicolor* and the *virginica*. This dataset was originally introduced in 1936 by Ronald Fisher. Using the various features of the flower (independent variables), we have to classify a given flower using Naive Bayes Classification model.

Step 1: Importing the Libraries

As always, the first step will always include importing the libraries which are the NumPy, Pandas and the Matplotlib.

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

Step 2: Importing the dataset

In this step, we shall import the Iris Flower dataset which is stored in my github repository as IrisDataset.csv and save it to the variable dataset. After this, we assign the 4 independent variables to ***X*** and the dependent variable 'species' to ***Y***. The first 5 rows of the dataset are displayed.

```
dataset =
pd.read_csv('https://raw.githubusercontent.com/mk-gurucharan/Classification/master/IrisDataset.csv')X =
dataset.iloc[:,:4].values
y = dataset['species'].valuesdataset.head(5)>>
sepal_length sepal_width petal_length petal_width species
5.1    3.5          1.4          0.2          setosa
4.9    3.0          1.4          0.2          setosa
4.7    3.2          1.3          0.2          setosa
4.6    3.1          1.5          0.2          setosa
5.0    3.6          1.4          0.2          setosa
```

Step 3: Splitting the dataset into the Training set and Test set

Once we have obtained our data set, we have to split the data into the training set and the test set. In this data set, there are 150 rows with 50 rows of each of the 3 classes. As each class is given in a continuous order, we need to randomly split the dataset. Here, we have the test_size=0.2, which means that ***20%*** of the dataset will be used for testing purposes as the ***test set*** and the remaining ***80%*** will be used as the ***training set*** for training the Naive Bayes classification model.

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2)
```

Step 4: Feature Scaling

The dataset is scaled down to a smaller range using the Feature Scaling option. In this, both the X_train and X_test values are scaled down to smaller values to improve the speed of the program.

```
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
```

Step 5: Training the Naive Bayes Classification model on the Training Set

In this step, we introduce the class GaussianNB that is used from the sklearn.naive_bayes library. Here, we have used a Gaussian model, there are several other models such as Bernoulli, Categorical and Multinomial. Here, we assign the GaussianNB class to the variable classifier and fit the X_train and y_train values to it for training purpose.

```
from sklearn.naive_bayes import GaussianNB
classifier = GaussianNB()
classifier.fit(X_train, y_train)
```

Step 6: Predicting the Test set results

Once the model is trained, we use the the classifier.predict() to predict the values for the Test set and the values predicted are stored to the variable y_pred.

```
y_pred = classifier.predict(X_test)
y_pred
```

Step 7: Confusion Matrix and Accuracy

This is a step that is mostly used in classification techniques. In this, we see the Accuracy of the trained model and plot the confusion matrix.

The confusion matrix is a table that is used to show the number of correct and incorrect predictions on a classification problem when the real values of the Test Set are known. It is of the format

True Positive	False Positive
False Negative	True Negative

The True values are the number of correct predictions made.

```
from sklearn.metrics import confusion_matrix
cm = confusion_matrix(y_test, y_pred)from sklearn.metrics import accuracy_score
print ("Accuracy : ", accuracy_score(y_test, y_pred))
cm>>Accuracy : 0.9666666666666667>>array([[14, 0, 0],
```

```
[ 0,  7,  0],
[ 0,  1,  8]])
```

From the above confusion matrix, we infer that, out of 30 test set data, 29 were correctly classified and only 1 was incorrectly classified. This gives us a high accuracy of 96.67%.

Step 8: Comparing the Real Values with Predicted Values

In this step, a Pandas DataFrame is created to compare the classified values of both the original Test set (***y_test***) and the predicted results (***y_pred***).

```
df = pd.DataFrame({'Real Values':y_test, 'Predicted Values':y_pred})
df>>
Real Values Predicted Values
setosa  setosa
setosa  setosa
virginica
virginica versicolor
versicolor setosa
setosa
setosa  setosa
... ... ... ... ...
virginica
versicolor virginica
virginica setosa
setosa
setosa  setosa
versicolor
versicolor versicolor
versicolor
```

This step is an additional step which is not much informative as the Confusion matrix and is mainly used in regression to check the accuracy of the predicted value.

CONCLUSION: Thus we have applied naïve bayes algorithm for social data analysis

FAQs:

1.What is the basic principle for applying naïve bayes algorithm for data analysis
2.write advantages and disadvantages of Naïve Bayes.

3. what are the desired characteristics of data before applying Naïve Bayes .
4. What is a confusion matrix?
5. Explain TP,FP, TN and FN.

Group A
Assignment No. 7

TITLE: Text Analysis for POS tagging, TF-IDF.

PROBLEM STATEMENT:
1. Extract Sample document and apply following document preprocessing methods:Tokenization, POS Tagging, stop words removal, Stemming and Lemmatization.
2. Create representation of document by calculating Term Frequency and Inverse Document Frequency.

COURSE OBJECTIVES: To develop in depth understanding and implementation of the key technologies in Data Science and Big Data Analytics

COURSE OUTCOMES:
Implement and evaluate data analytics algorithms

THEORY:

Text mining is also referred to as text analytics. Text mining is a process of exploring sizeable textual data and find patterns. Text Mining process the text itself, while NLP process with the underlying metadata. Finding frequency counts of words, length of the sentence, presence/absence of specific words is known as text mining. Natural language processing is one of the components of text mining. NLP helps identified sentiment, finding entities in the sentence, and category of blog/article. Text mining is preprocessed data for text analytics. In Text Analytics, statistical and machine learning algorithm used to classify information.

Text Analysis Operations using NLTK

NLTK is a powerful Python package that provides a set of diverse natural languages algorithms. It is free, opensource, easy to use, large community, and well documented. NLTK consists of the most common algorithms such as tokenizing, part-of-speech tagging, stemming, sentiment analysis, topic segmentation, and named entity recognition. NLTK helps the computer to analysis, preprocess, and understand the written text.

Tokenization

Tokenization is the first step in text analytics. The process of breaking down a text paragraph into smaller chunks such as words or sentence is called Tokenization. Token is a single entity that is building blocks for sentence or paragraph.

Sentence Tokenization

Sentence tokenizer breaks text paragraph into sentences.

Word Tokenization

Word tokenizer breaks text paragraph into words.

Stopwords

Stopwords considered as noise in the text. Text may contain stop words such as is, am, are, this, a, an, the, etc.

In NLTK for removing stopwords, you need to create a list of stopwords and filter out your list of tokens from these words.

Lexicon Normalization

Lexicon normalization considers another type of noise in the text. For example, connection, connected, connecting word reduce to a common word "connect". It reduces derivationally related forms of a word to a common root word.

Stemming

Stemming is a process of linguistic normalization, which reduces words to their word root word or chops off the derivational affixes. For example, connection, connected, connecting word reduce to a common word "connect".

Lemmatization

Lemmatization reduces words to their base word, which is linguistically correct lemmas. It transforms root word with the use of vocabulary and morphological analysis. Lemmatization is usually more sophisticated than stemming. Stemmer works on an individual word without knowledge of the context. For example, The word "better" has "good" as its lemma. This thing will miss by stemming because it requires a dictionary look-up.

POS Tagging

The primary target of Part-of-Speech(POS) tagging is to identify the grammatical group of a given word. Whether it is a NOUN, PRONOUN, ADJECTIVE, VERB, ADVERBS, etc. based on the context. POS Tagging looks for relationships within the sentence and assigns a corresponding tag to the word.

CONCLUSION:

Thus NLTK can be used for Text analysis , text preprocessing, POS tagging etc.

FAQs:

1. What is Text Analysis.
2. Explain any two applications where text analysis is used.
3. What is text preprocessing.
4. What is POS tagging
5. What is TF/IDF?

Group A
Assignment No. 8

TITLE: Data Visualization -I

PROBLEM STATEMENT: . Use the inbuilt dataset 'titanic'. The dataset contains 891 rows and contains information about the passengers who boarded the unfortunate Titanic ship.
1. Use the Seaborn library to see if we can find any patterns in the data.
2. Write a code to check how the price of the ticket (column name: 'fare') for each passenger is distributed by plotting a histogram.

COURSE OBJECTIVES: To gain practical, hands-on experience with statistics programming languages and Big Data tools

COURSE OUTCOMES: Students will be able to gain basic knowledge of different data visualization techniques.

THEORY:

Data visualization :

It is the practice of translating information into a visual context, such as a map or graph, to make data easier for the human brain to understand and pull insights from. The main goal of data visualization is to make it easier to identify patterns, trends and outliers in large data sets.

ADVANTAGES:

- Free Guide: 10 Worthy Future Trends in DataEasy to understand.
- Spot patterns.
- Story telling is more engaging.
- Speed up the decision-making process.

Seaborn:

Seaborn is an open-source Python library built on top of **matplotlib**. It is used for data visualization and exploratory data analysis. Seaborn works easily with dataframes and the Pandas library. The graphs created can also be customized easily. Below are a few benefits of Data Visualization.

A histogram is basically used to represent data provided in a form of some groups.It is accurate method for the graphical representation of numerical data distribution.It is a type of bar plot where X-axis represents the bin ranges while Y-axis gives information about frequency.

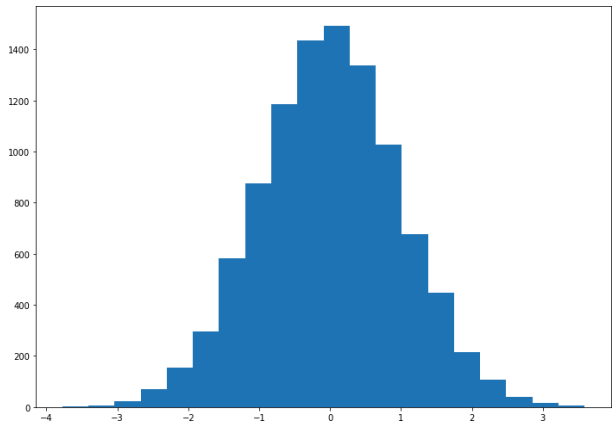
To create a histogram the first step is to create bin of the ranges, then distribute the whole range of the values into a series of intervals, and count the values which fall into each of the intervals.Bins are clearly identified as consecutive, non-overlapping intervals of variables.The matplotlib.pyplot.hist() function is used to compute and create histogram of x.

The following table shows the parameters accepted by matplotlib.pyplot.hist() function :

Attribute	parameter
x	array or sequence of array
bins	optional parameter contains integer or sequence or strings
density	optional parameter contains boolean values
range	optional parameter represents upper and lower range of bins
histtype	optional parameter used to create type of histogram [bar, barstacked, step, stepfilled], default is “bar”
align	optional parameter controls the plotting of histogram [left, right, mid]

Customization of Histogram

Matplotlib provides a range of different methods to customize histogram. matplotlib.pyplot.hist() function itself provides many attributes with the help of which we can modify a histogram.The hist() function provide a patches object which gives access to the properties of the created objects, using this we can modify the plot according to our will.



OUTPUT:

CONCLUSION: Students will design and develop big data solutions using different data visualization techniques

FAQs:

1. How Data visualization helps in understanding data characteristics.
2. What is Histogram?
3. Write any 3 imp. Functions provided in seaborn library.
4. What is data visualization?
5. What is a box plot?

Group A
Assignment No. 9

TITLE: Data Visualization -II

PROBLEM STATEMENT: . Use the inbuilt dataset 'titanic' as used in the above problem.
1. Plot a box plot for distribution of age with respect to each gender along with the information about whether they survived or not. (Column names : 'sex' and 'age')
2. Write observations on the inference from the above statistics.

COURSE OBJECTIVES: To gain practical, hands-on experience with statistics programming languages and Big Data tools

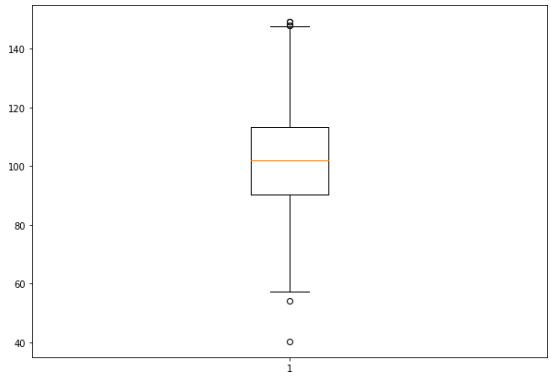
COURSE OUTCOMES: Students will be able to gain basic knowledge of different data visualization techniques.

THEORY:

Box Plot:
A **Box Plot** is also known as **Whisker plot** is created to display the summary of the set of data values having properties like minimum, first quartile, median, third quartile and maximum. In the box plot, a box is created from the first quartile to the third quartile, a vertical line is also there which goes through the box at the median. Here x-axis denotes the data to be plotted while the y-axis shows the frequency distribution. The matplotlib.pyplot module of matplotlib library provides boxplot() function with the help of which we can create box plots.
Parameters:

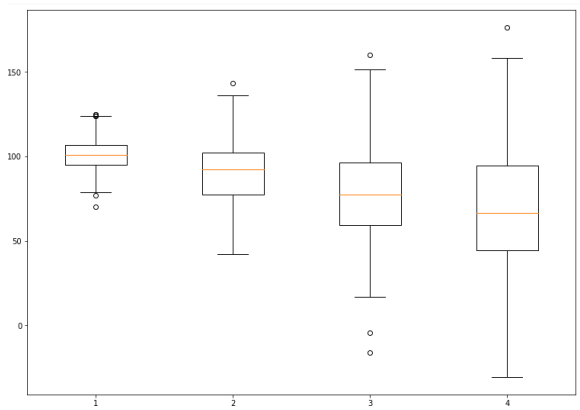
Attribute	Value
data	array or sequence of array to be plotted
notch	optional parameter accepts boolean values
vert	optional parameter accepts boolean values false and true for horizontal and vertical plot respectively
bootstrap	optional parameter accepts int specifies intervals around notched boxplots
usermedians	optional parameter accepts array or sequence of array dimension compatible with
data positions	optional parameter accepts array and sets the position of boxes
widths	optional parameter accepts array and sets the width of boxes

Attribute	Value
patch_artist	optional parameter having boolean values
labels	sequence of strings sets label for each
dataset	
meanline	optional having boolean value try to render meanline as full width of
box order	optional parameter sets the order of the boxplot



Customizing Box Plot

The matplotlib.pyplot.boxplot() provides endless customization possibilities to the box plot. The notch = True attribute creates the notch format to the box plot, patch_artist = True fills the boxplot with colors, we can set different colors to different boxes.The vert = 0 attribute creates horizontal box plot. labels takes same dimensions as the number data sets.



OUTPUT:

CONCLUSION: Students will design and develop big data solutions using different data visualization techniques

FAQs:

1. What types of data can you use with Tableau?
2. What is a scatter plot and when would you use one?
3. Explain when you might use a bar graph over a pie chart?
4. What is data validation?
5. Name several data validation techniques.

Group A
Assignment No. 10

TITLE: Data Visualization -III

PROBLEM STATEMENT: Download the Iris flower dataset or any other dataset into a DataFrame. (e.g.https://archive.ics.uci.edu/ml/datasets/Iris). Scan the dataset and give the inference as:
1. List down the features and their types (e.g., numeric, nominal) available in the dataset.
2. Create a histogram for each feature in the dataset to illustrate the feature distributions.
3. Create a box plot for each feature in the dataset.
4. Compare distributions and identify outliers.

COURSE OBJECTIVES: To gain practical, hands-on experience with statistics programming languages and Big Data tools

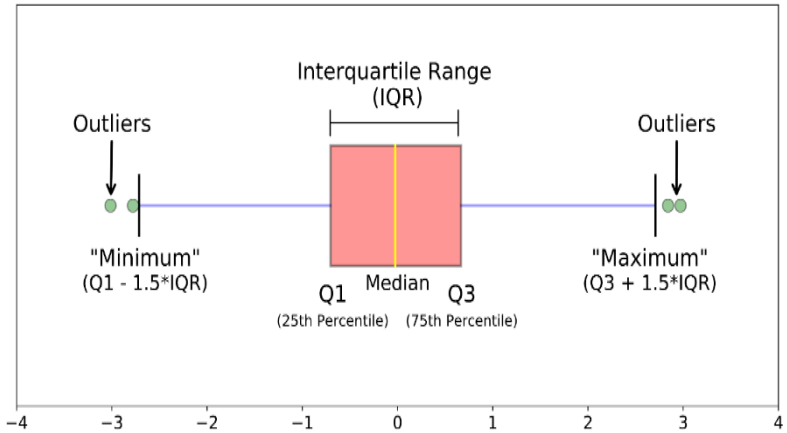
COURSE OUTCOMES: Students will be able to gain basic knowledge of different data visualization techniques.

THEORY:

OUTLIERS:

Outliers are an important part of a dataset. They can hold useful information about your data.Outliers can give helpful insights into the data you're studying, and they can have an effect on statistical results. This can potentially help you discover inconsistencies and detect any errors in your statistical processes.So, knowing how to find outliers in a dataset will help you better understand your data.In simple terms, an outlier is an extremely high or extremely low data point relative to the nearest data point and the rest of the neighboring co-existing values in a data graph or dataset you're working with.Outliers are extreme values that stand out greatly from the overall pattern of values in a dataset or graph.

When reviewing a box plot, an outlier is defined as a data point that is located outside the whiskers of the box plot.



For example, outside 1.5 times the interquartile range above the upper quartile and below the lower quartile

(Q1 - 1.5 * IQR or Q3 + 1.5 * IQR).

Adding a boxplot on top of a histogram can help you in better understanding the distribution of the data and visualizing outliers as well as quartiles positions. Note that we chose to use the histplot function below, though you could definitely use the distplot function also.

OUTPUT:

CONCLUSION: Students will design and develop big data solutions using different data visualization techniques

FAQs:

1. What are the data types in data visualization?
2. What are the data types in data visualization?
3. What is depth cueing in visualisation?
4. How can we statistically analyze and visualize data from different experiments?

Group B
Assignment No. 1

AIM: Hadoop Programming: Word Count MapReduce Program Using Eclipse

TITLE: Write a code in JAVA for a simple Word Count application that counts the number of occurrences of each word in a given input set using the Hadoop Map-Reduce framework on local-standalone set-up.

COURSE OBJECTIVES: To gain practical, hands-on experience with statistics programming languages and Big Data tools

COURSE OUTCOMES: Students will be able to use cutting edge tools and technologies to analyze Bid Data.

REQUIREMENTS:
Hadoop and MapReduce

THEORY:
In Hadoop, MapReduce is a computation that decomposes large manipulation jobs into individual tasks that can be executed in parallel across a cluster of servers. The results of tasks can be joined together to compute final results.

MapReduce consists of 2 steps:

- **Map Function –** It takes a set of data and converts it into another set of data, where individual elements are broken down into tuples (Key-Value pair).

  Example – (Map function in Word Count)

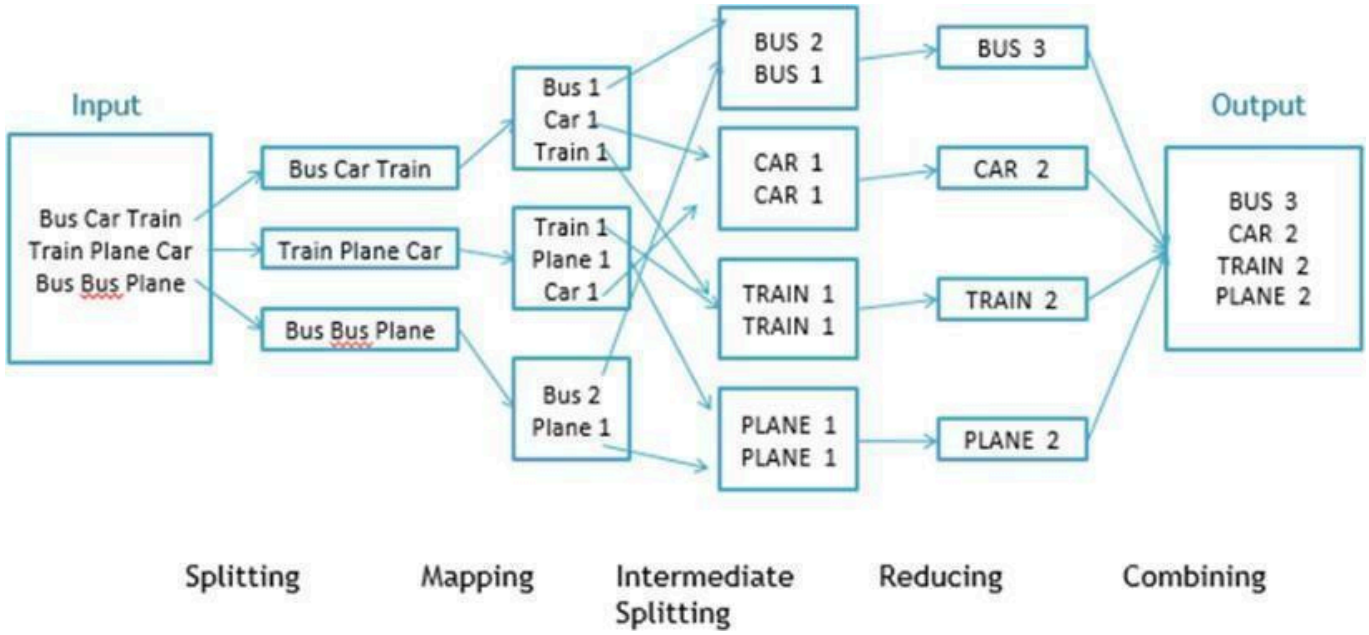
Input	Set of data	Bus, Car, bus, car, train, car, bus, car, train, bus, TRAIN,BUS, buS, caR, CAR, car, BUS, TRAIN
Output	Convert into another set of data (Key,Value)	(Bus,1), (Car,1), (bus,1), (car,1), (train,1), (car,1), (bus,1), (car,1), (train,1), (bus,1), (TRAIN,1),(BUS,1), (buS,1), (caR,1), (CAR,1), (car,1), (BUS,1), (TRAIN,1)

- **Reduce Function –** Takes the output from Map as an input and combines those data tuples into a smaller set of tuples.

Example – (Reduce function in Word Count)

Input **(output of Map function)**	Set of Tuples	(Bus,1), (Car,1), (bus,1), (car,1), (train,1), (car,1), (bus,1), (car,1), (train,1), (bus,1), (TRAIN,1),(BUS,1), (buS,1), (caR,1), (CAR,1), (car,1), (BUS,1), (TRAIN,1)
Output	Converts into smaller set of tuples	(BUS,7), (CAR,7), (TRAIN,4)

Work Flow of the Program



Workflow of MapReduce consists of 5 steps:

1. **Splitting** – The splitting parameter can be anything, e.g. splitting by space, comma, semicolon, or even by a new line ('\n').
2. **Mapping** – as explained above.

3. **Intermediate splitting** – the entire process in parallel on different clusters. In order to group them in "Reduce Phase" the similar KEY data should be on the same cluster.
4. **Reduce** – it is nothing but mostly group by phase.
5. **Combining** – The last phase where all the data (individual result set from each cluster) is combined together to form a result.

Counting the number of words in any language is a piece of cake like in C, C++, Python, Java, etc. MapReduce also uses Java but it is very easy if you know the syntax on how to write it. It is the basic of MapReduce. You will first learn how to execute this code similar to "Hello World" program in other languages. So here are the steps which show how to write a MapReduce code for Word Count. Steps:

1. Open Eclipse> File > New > Java Project >(Name it – MRProgramsDemo) > Finish.

Create Three Java Classes into the project. Name them WCDriver(having the main function), WCMapper, WCReducer.
You have to include two Reference Libraries for that:
Right Click on Project -> then select Build Path-> Click on Configure Build Path

In the above figure, you can see the Add External JARs option on the Right Hand Side. Click on it and add the below mentioned files. You can find these files in /usr/lib/
1. /usr/lib/hadoop-0.20-mapreduce/hadoop-core-2.6.0-mr1-cdh5.13.0.jar
2. /usr/lib/hadoop/hadoop-common-2.6.0-cdh5.13

.0.jar Copy mapper code

Mapper code:

```
// Importing libraries
import
java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.Mapper;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reporter;
public class WCMapper extends MapReduceBase implements Mapper&lt;LongWritable,
Text, Text, IntWritable&gt; {
// Map function
public void map(LongWritable key, Text value, OutputCollector&lt;Text,
```

```
IntWritable&gt; output, Reporter rep) throws IOException
{
String line = value.toString();
// Splitting the line on spaces
for (String word : line.split(&quot; &quot;))
{
if (word.length() &gt; 0)
{
output.collect(new Text(word), new IntWritable(1));
}
}
}
}
```

Copy reducer code

Reducer code:

```
//    Importing        libraries
import
java.io.IOException;
import java.util.Iterator;
import    org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import
org.apache.hadoop.mapred.MapReduceBase;
import  org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reducer;
import org.apache.hadoop.mapred.Reporter;
public class WCReducer extends MapReduceBase implements Reducer&lt;Text,
IntWritable, Text, IntWritable&gt; {

// Reduce function
public        void        reduce(Text        key,        Iterator&lt;IntWritable&gt;        value,
OutputCollector&lt;Text, IntWritable&gt; output,
Reporter rep) throws IOException
{
int count = 0;
// Counting the frequency of each words
while (value.hasNext())
{
IntWritable   i   =   value.next();
count += i.get();
}
output.collect(key, new IntWritable(count));
}
}
```

Copy driver code

```
// Importing libraries
import java.io.IOException;
import
org.apache.hadoop.conf.Configured;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
```

Conclusion – Thus we created a word count program using Hadoop map-reduce.

FAQ

1. What is Hadoop? What are the different components of the Hadoop Ecosystem?
2. List different advantages and disadvantages of Hadoop.
3. What is HDFS? What is the purpose of it?
4. What are the different Hadoop configuration files?
5. What are the differences between regular FileSystem and HDFS?

Group B
Assignment 2

TITLE: Design of Distributed Application -Processing Log file using Map-Reduce Framework

PROBLEM STATEMENT: Design a distributed application using Map-Reduce which processes a log file of a system.

COURSE OBJECTIVES: To gain practical, hands-on experience with Big Data tools

COURSE OUTCOMES: Students will be able to use cutting edge tools and technologies to analyze Bid Data.

THEORY:

What is MapReduce?

MapReduce is a processing technique and a program model for distributed computing based on java. The MapReduce algorithm contains two important tasks, namely Map and Reduce. Map takes a set of data and converts it into another set of data, where individual elements are broken down into tuples (key/value pairs). Secondly, reduce task, which takes the output from a map as an input and combines those data tuples into a smaller set of tuples. As the sequence of the name MapReduce implies, the reduce task is always performed after the map job.

The Algorithm

MapReduce program executes in three stages, namely map stage, shuffle stage, and reduce stage.

- Map stage – The map or mapper's job is to process the input data. Generally, the input data is in the form of file or directory and is stored in the Hadoop file system (HDFS). The input file is passed to the mapper function line by line. The mapper processes the data and creates several small chunks of data.
- Reduce stage – This stage is the combination of the Shuffle stage and the Reduce stage. The Reducer's job is to process the data that comes from the mapper. After processing, it produces a new set of output, which will be stored in the HDFS.

- Most of the computing takes place on nodes with data on local disks that reduces the network traffic.

Inputs and Outputs (Java Perspective)

The MapReduce framework operates on <key, value> pairs, that is, the framework views the input to the job as a set of <key, value> pairs and produces a set of <key, value> pairs as the output of the job,

conceivably of different types.

The key and the value classes should be serialized by the framework and hence, need to implement the Writable interface. Additionally, the key classes have to implement the Writable-Comparable interface to facilitate sorting by the framework. Input and Output types of a MapReduce job – (Input) <kl, vl> → map
→ <k2, v2> → reduce → <k3, v3>(Output).

The input for our program is txt_file .

1.Steps for Compilation & Execution of Program:

#sudo mkdir analyzelogs

ls

#sudo chmod -R 777 analyzelogs/

cd

ls

cd

..

pwd ls cd

pwd

#sudo chown -R hduser analyzelogs/

cd ls

#cd analyzelogs/

ls cd ..

2.Copy the Files (Mapper.java,Reduce.java,Driver.java to Analyzelogs Folder)

#sudo cp /home/isha/Desktop/count_logged_users/* -/analyzelogs/

Start HADOOP

#start-dfs.sh

#start-yarn.sh

#jps

3. Compile Java Files

javac -d .

DsbdaMapper.java DsbdaReducer.java DsbdaDriver.java ls

#cd Dsbda/

ls cd .. #sudo gedit Manifest.txt

#jar -cfm analyzelogs.jar Manifest.txt Dsbda/*.class

ls cd jps

#cd analyzelogs/

4. Create Directory on Hadoop #sudo mkdir ~/input2000

ls pwd #sudo cp access_log_short.csv ~/input2000/

$HADOOP_HOME/bin/hdfs dfs -put ~/input2000 /

$HADOOP_HOME/bin/hadoop jar analyzelogs.jar /input2000 /output2000

$HADOOP_HOME/bin/hdfs dfs -cat /output2000/part-00000

stop-all.sh

jps

CONCLUSION: Thus, we have learnt how to design a distributed application using MapReduce and process a log file of a system

FAQs

1. What is Hadoop? What are the different components of the Hadoop Ecosystem?
2. List different advantages and disadvantages of Hadoop.
3. What is HDFS? What is the purpose of it?
4. What are the different Hadoop configuration files?
5. What are the differences between regular FileSystem and HDFS?

Group B
Assignment No. 3

TITLE: Weather Data Forecasting – Map Reduce

PROBLEM STATEMENT: Locate dataset (e.g., sample_weather.txt) for working on weather data which reads the text input files and finds average for temperature, dew point and wind speed.

COURSE OBJECTIVES:
To gain practical, hands-on experience with statistics programming languages and Big Data tools.

COURSE OUTCOMES: Students will be able to use cutting edge tools and technologies to analyze Bid Data.

THEORY:
Introduction to Map- Reduce:
MapReduce is a framework using which we can write applications to process huge amounts of data, in parallel, on large clusters of commodity hardware in a reliable manner. MapReduce is a processing technique and a program model for distributed computing based on java.
The major advantage of MapReduce is that it is easy to scale data processing over multiple computing nodes
Under the MapReduce model, the data processing primitives are called mappers and reducers. Decomposing a data processing application into mappers and reducers is sometimes nontrivial. But, once we write an application in the MapReduce form, scaling the application to run over hundreds, thousands, or even tens of thousands of machines in a cluster is merely a configuration change. This simple scalability is what has attracted many programmers to use the MapReduce model

The Algorithm
MapReduce program executes in three stages, namely map stage, shuffle stage, and reduce stage.

Mapstage : The map or mapper's job is to process the input data. Generally the input data is in the form of file or directory and is stored in the Hadoop file system (HDFS). The input file is passed to the mapper function line by line. The mapper processes the data and creates several small chunks of data.

Reduce stage : This stage is the combination of the Shuffle stage and the Reduce stage. The Reducer's job is to process the data that comes from the mapper. After processing, it produces a new set of output, which will be stored in the HDFS.

Inserting Data into HDFS:
•The MapReduce framework operates on pairs, that is, the framework views the input to the job as a set of pairs and produces a set of pairs as the output of the job, conceivably of different types.

• The key and the value classes should be in serialized manner by the framework and hence, need to implement the Writable interface. Additionally, the key classes have to implement the Writable

-Comparable interface to facilitate sorting by the framework.

•Input and Output types of a MapReduce job: (Input) <k1,v1>-> map -><k1,v2> -> reduce -><k1,v3> (Output).

1. To start with Map reduce Project, please install Apache Hadoop , Refer the below site for more details
https://hadoop.apache.org/docs/current/hadoop-mapreduce-client/hadoop-mapreduce-client-core/MapReduc eTutorial.html

sample_whether.txt file will look like this

sample_weather.txt - Notepad

File Edit Format View Help

```
690190 13910  20060201_0  51.75    33.0 24  1006.3 24  943.9 24  15.0 24  10.7 24  22.0  28.9  0.00I 999.9  000000
690190 13910  20060201_1  54.74    33.0 24  1006.3 24  943.9 24  15.0 24  10.7 24  22.0  28.9  0.00I 999.9  000000
690190 13910  20060201_2  50.59    33.0 24  1006.3 24  943.9 24  15.0 24  10.7 24  22.0  28.9  0.00I 999.9  000000
690190 13910  20060201_3  51.67    33.0 24  1006.3 24  943.9 24  15.0 24  10.7 24  22.0  28.9  0.00I 999.9  000000
690190 13910  20060201_4  65.67    33.0 24  1006.3 24  943.9 24  15.0 24  10.7 24  22.0  28.9  0.00I 999.9  000000
690190 13910  20060201_5  55.37    33.0 24  1006.3 24  943.9 24  15.0 24  10.7 24  22.0  28.9  0.00I 999.9  000000
690190 13910  20060201_6  49.26    33.0 24  1006.3 24  943.9 24  15.0 24  10.7 24  22.0  28.9  0.00I 999.9  000000
690190 13910  20060201_7  55.44    33.0 24  1006.3 24  943.9 24  15.0 24  10.7 24  22.0  28.9  0.00I 999.9  000000
690190 13910  20060201_8  64.05    33.0 24  1006.3 24  943.9 24  15.0 24  10.7 24  22.0  28.9  0.00I 999.9  000000
690190 13910  20060201_9  68.77    33.0 24  1006.3 24  943.9 24  15.0 24  10.7 24  22.0  28.9  0.00I 999.9  000000
```

2. Install maven dependencies/plugins required for Hadoop
   $ mvn install

3. Add the sample input file given in the project to the hdfs hdfs dfs -put sample_weather.txt
4. Then, Start the Map reduce application to submit the job to Hadoop
   To run: bin/hadoop jar target/weather-1.0.jar [-m maps] [-rreduces] in-dir for job 1 out-dir for job 1 out-dir for job 2

OUTPUT:

CONCLUSION: Thus, we have learnt how to process input data using map reduce framework.

FAQs:

1. What is map-reduce?
2. Explain in how many stages the map-reduce program executes with example?
3. How to insert data into HDFS?
4. How to Interact with MapReduce Jobs?
5. Write Advantages of Hadoop MapReduce

Assignment No.14

Miniproject

Mini Project helps you to explore and strengthen the understanding of fundamentals through practical application of theoretical concepts. Mini Project can help you to boost your skills and widen your horizon of thinking. It acts like a beginners guide to do larger projects later in their career.

Students will perform miniproject individual.

Steps to carry miniproject are as follow-

1. Selection of Topic

Selection of topic is a huge and important task in a Mini Project. One should have a clear idea about one's subject strengths and the selected topic should be relevant to it. Always select the project that has value addition.

As a graduate you should select a project which is either advantageous to a lot of people or enhance your technical and managerial skills. Your project must play its role towards a positive growth/development in that specific field.

2. Research about the selected topic online

Do some online research about the selected topic. Go through the research papers from different researchers around the world on the topics related to Mini Project. Find some websites containing the information about the materials used for Mini Project.

3. Suggestions from subject experts

Go to the subject experts in your college and interact with them about the Mini Project topic. You can also meet many subject experts from various parts of India through social media and some discussion forums. This helps you in getting suggestions in different possible ways, through which you can get a clear idea on your Mini Project topic.

4. Planning

After getting a clear idea about the topic, prepare a rough plan about procurement of resources, experimentation and fabrication along with your teammates. Make a rough schedule, adapt to it and distribute the work among your teammates. This will keep your Mini Project on track and individuals will come to know about their part in the Mini Project rather than any individual (leader) taking full responsibilities.

5. Execution of plans

Make sure that the materials will be ready for the experimentation/fabrication by the scheduled time. Follow the schedule during experimentation/fabrication to get accurate and efficient results.

6. Presentation

Experimentation/Fabrication does not make a Mini Project successful; one should be able to present the results in

proper way. So it should be prepared in such a way that, it reflects the exact objective of your Mini Project.

Some of the key features of drafting good mini projects report are:

- Facilitates quicker and easier way to communicate the information
- Can assist in selective reading
- Easier navigation to contents with numbered headings and sub headings
- Better explanations with figures, tables and charts

Marathwada Mitra Mandal's
College of Engineering, Karvenagar, Pune
Department of Computer Engineering

MINI PROJECT FORMAT

- Title:
- Abstract:
- Introduction:
- Problem Statement:
- Technical
    - Software (Front, Back):
    - Dataset (Link, Metadata):
    - Library:
    - Functions Used:
- Algorithm (type of algorithm):
- Performance metrics:
- Results/ Visualizations:
- References:

Assignment No. 15

VLab assignment

Link - https://nlp-iiith.vlabs.ac.in/exp/word-analysis/

Aim- A word can be simple or complex. For example, the word 'cat' is simple because one cannot further decompose the word into smaller part. On the other hand, the word 'cats' is complex, because the word is made up of two parts: root 'cat' and plural suffix '-s'

Theory –
Analysis of a word into root and affix(es) is called Morphological analysis of a word. It is mandatory to identify the root of a word for any natural language processing task. A root word can have various forms. Indian languages are generally morphologically rich languages and therefore morphological analysis of words becomes a very significant task for Indian languages.

Types of Morphology

Morphology is of two types,

Inflectional morphology
Deals with word forms of a root, where there is no change in lexical category. For example, 'played' is an inflection of the root word 'play'. Here, both 'played' and 'play' are verbs.

Derivational morphology
Deals with word forms of a root, where there is a change in the lexical category. For example, the word form 'happiness' is a derivation of the word 'happy'. Here, 'happiness' is a derived noun form of the adjective 'happy'.

Morphological Features:

All words will have their lexical category attested during morphological analysis. A noun and pronoun can take suffixes of the following features: gender, number, person, case.A verb can take suffixes of the following features: tense, aspect, modality, gender, number, person.
The value of tense can be present, past or future. This feature is applicable for verbs. The value of aspect can be perfect (pft), continuous (cont) or habitual (hab). This feature is not applicable for verbs.

'case' can be direct or oblique. This feature is applicable for nouns. A case is an oblique case when a postposition occurs after noun. If no postposition can occur after noun, then the case is a direct case. This is applicable for hindi but not english as it doesn't have any postpositions.

Objectives –

The objective of the experiment is to learn about morphological features of a word by analysing it.

Procedure-

STEP 1: Select the language.

OUTPUT: Drop down for selecting words will appear.

STEP 2: Select the word.

OUTPUT: Drop down for selecting features will appear.

STEP 3: Select the features.

STEP 4: Click "Check" button to check your answer.

OUTPUT: Right features are marked by tick and wrong features are marked by cross.

FAQ-

1. What is morphological analysis?
2. Why is identifying the root of a word important for natural language processing (NLP) tasks?.
3. Can a root word have various forms?
4. Why is morphological analysis particularly significant for Indian languages?
5. What are some challenges in morphological analysis of Indian languages?

References-

1. Speech and Language Processing - An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition BY: Daniel Jurafsky and James H. Martin - Chapter 3

2. Natural Language Processing - A Paninian Perspective BY: Akshar Bharti, Vineet Chaitanya and Rajeev Sangal - Chapter 3